MATHEMATICS

COMPLEX NUMBER

THEORY SHEET

ENGLISH

MATRIX

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1.1 INTRODUCTION TO COMPLEX NUMBER

Complex numbers are defined as expressions of the form $a + ib$ where $a, b \in R$ & $i = \sqrt{-1}$.

It is denoted by z i.e. $z = a + ib$.

'a' is called real part of z (Re z) and 'b' is called imaginary part of z (Im z).

$z = a + ib$

purely real	purely imaginary	imaginary
if $b = 0$	if $a = 0$	if $b \neq 0$

Note : (a) The set R of real numbers is a proper subset of the Complex Numbers. Hence the complete number system is $N \subset W \subset I \subset Q \subset R \subset C$.

(b) Zero is purely real as well as purely imaginary but not imaginary.

1.2 INTEGRAL POWERS OF IOTA

$i = \sqrt{-1}$ so $i^2 = -1$

$i^3 = -i$ and $i^4 = 1$

$i^{4n} = 1$ $\quad$ $i^{4n+1} = i$,

$i^{4n+2} = -1$ $\quad$ $i^{4n+3} = -i$,

Q.1 Find the value of $[i]^{198}$

Ans. –1

Sol. $[i]^{198} = [i^2]^{99} = [-1]^{99} = -1$

Q.2 Find the value of $i^n + i^{n+1} + i^{n+2} + i^{n+3}$

Ans. 0

Sol. $i^n + i^{n+1} + i^{n+2} + i^{n+3}$

$= i^n [1 + i + i^2 + i^3]$

$= i^n [1 + i - 1 - i] = i^n [0] = 0$

Q.3 Find $1 + i + i^2 + \ldots\ldots + i^{n-1}$

Ans.

Sol. sum = 0 if $n = 4k$

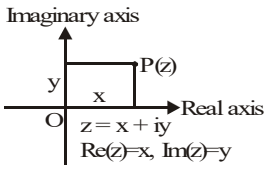
sum = 1 if $n = 4k + 1$

sum = 1 + i if $n = 4k + 2$

sum = i if $n = 4k + 3$

1.3 ARGAND PLANE / COMPLEX PLANE

Master Argand had done a systematic study on complex numbers and represented every complex number $z = x + iy$ as a set of ordered pair (x, y) on a plane called complex plane (Argand Diagram) containing two perpendicular axes. Horizontal axis is known as Real axis & vertical axis is known as Imaginary axis.



Note:

1. All complex numbers lying on the real axis are called as purely real
2. All complex numbers lying on imaginary axis are purely imaginary.

1.4 ALGEBRA OF COMPLEX NUMBER

In performing operations with complex numbers we can proceed as in the algebra of real numbers, replacing i^2 by -1 when it occurs.

1. Addition $(a + bi) + (c + di) = a + bi + c + di = (a + c) + (b + d)\, i$
   Subtraction $(a + bi) - (c + di) = a + bi - c - di = (a - c) + (b - d)\, i$
2. Multiplication $(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$
3. Division $$\frac{a+bi}{c+di} = \frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac - adi + bci - bdi^2}{c^2 - d^2 i^2}$$
   $$= \frac{ac + bd + (bc - ad)i}{c^2 + d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}\, i$$
4. Two complex numbers z_1 and z_2 are said to be equal if $Re(z_1) = Re(z_2)$ and $Im(z_1) = Im(z_2)$ (i.e. they occupy the same position on complex plane)
5. If $z_1 z_2 = 0 \Rightarrow z_1 = 0$ or $z_2 = 0$
6. Inequalities in imaginary numbers are not defined.
   If $a + ib > c + id$ has to be meaningful $\Rightarrow$ $b = d = 0$. **e.g.** $4 + 2i < 2 + 4i$ are meaningless.
7. In real numbers if $a^2 + b^2 = 0$ then $a = 0 = b$ however in complex numbers, $z_1^2 + z_2^2 = 0$ does not imply $z_1 = z_2 = 0$.

1.5 SQUARE ROOT OF A COMPLEX NUMBER

Find square root of $9 + 40i$

Let $x + iy = \sqrt{9 + 40i}$

$(x + iy)^2 = 9 + 40i$

$\therefore \quad x^2 - y^2 = 9 \quad$ .....(i)

and $\quad xy = 20 \quad$ .....(ii)

squaring (i) and adding with 4 times the square of (ii)

we get $x^4 + y^4 - 2x^2 y^2 + 4x^2 y^2 = 81 + 1600$

$\Rightarrow \quad (x^2 + y^2)^2 = 1681$

$\Rightarrow \quad x^2 + y^2 = 41 \quad$ .....(iii)

from (i) + (iii) we get

$x^2 = 25 \quad \Rightarrow \quad x = \pm 5$

$y^2 = 16 \quad \Rightarrow \quad y = \pm 4$

from equation (ii) we can see that

x & y are of same sign

$\therefore \quad x + iy = (5 + 4i)$ or $-(5 + 4i)$

$\therefore \quad$ sq. roots of $9 + 40i = \pm (5 + 4i)$

Ans. $\pm (5 + 4i)$

2.1 CONJUGATE AND ITS PROPERTIES

The complex numbers $z = (a, b) = a + ib$ and $\bar{z} = (a, -b) = a - ib$ where $b \neq 0$ are said to be complex conjugate of each other (Here the complex conjugate is obtained by just changing the sign of i)

z and $\bar{z}$ are image of each other w.r.t. real axis.

Properties of Conjugate Complex Number

Let $z = a + ib$ and $\bar{z} = a - ib$ then

1. $\overline{(\bar{z})} = z$ 2. $z + \bar{z} = 2\,\text{Re}(z)$; $z - \bar{z} = = 2i\,\text{Im}(z)$

3. $z = \bar{z} \Rightarrow$ z is purely real

   $z + \bar{z} = 0$ or $z = -\bar{z} \Rightarrow$ z is purely imaginary

4. $\overline{z_1+z_2} = \overline{z_1}+\overline{z_2}$; $\overline{z_1-z_2} = \overline{z_1} - \overline{z_2}$

5. $\overline{z_1 z_2} = \overline{z_1}\ \overline{z_2}$; $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$; $\overline{z^n} = (\bar{z})^n$

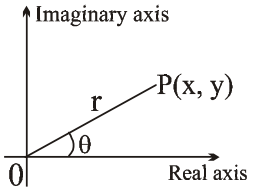
6. $z_1 = z_2 \Rightarrow \overline{z_1} = \overline{z_2}$

Q.1 Prove that $z_1\overline{z_2} + z_2\overline{z_1}$ is purely real.

2.2 MODULUS

If P denotes complex number z = x + iy, then the length OP is called modulus of complex number z. It is denoted by |z|.

$OP = |z| = \sqrt{x^2 + y^2}$



Geometrically $|z|$ represents the distance of point P from origin. ($|z| \geq 0$)

Note : Unlike real numbers, $|z| = \begin{cases} z & \text{if } z > 0 \\ -z & \text{if } z < 0 \end{cases}$ is not correct.

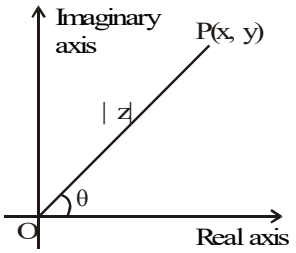
Properties of modulus of a Complex Number

1. $|z| \geq 0$ 2. $\text{Re}(z) \leq |z|$; $\text{Im}(z) \leq |z|$
3. $|z| = |\bar{z}| = |-z| = |-\bar{z}|$ 4. $z\,\bar{z} = |z|^2$
5. If $z = a + ib$ then $|z|^2 = a^2 + b^2$ & $z^2 = a^2 - b^2 + 2iab$
6. $|z_1 + z_2| \neq |z_1| + |z_2|$

   Ex. - $z_1 = 3 + 4i$, $z_2 = 3 - 4i$, LHS = 6, RHS = 5 + 5
7. $|z_1\ z_2| = |z_1|\ |z_2|$; $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ $(z_2 \neq 0)$; $|z|^n = |z^n|$, $n \in N$
8. If $z_1 = z_2 \Rightarrow |z_1| = |z_2|$ But converse is not true

2.3 ARGUMENT AND ITS PROPERTIES

If P denotes complex number z = x + iy and if OP makes an angle θ with +ve real axis, then θ is called one of the arguments of z.



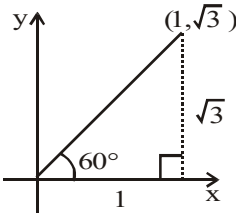
Note :

Argument of a complex number is a many valued function. If θ is the argument of a complex number, then $2n\pi + \theta$; $n \in I$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$.

Q.1 For $z = 1 + i\sqrt{3}$

$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$

$\arg(z) = 2n\pi + \frac{\pi}{3}, \quad n \in I$



Amplitude (principal value of the argument)

The unique value of θ such that $-\pi < \theta \le \pi$ is called Amplitude (principal value of the argument).

Unless otherwise stated, amp z implies principal value of the argument.

Least positive argument

The unique value of θ such that $0 < \theta \le 2\pi$ is called **least positive argument.**

Note:

1. If $z = 0$, arg(z) is not defined
2. If z is real & negative, $\arg(z) = \pi$.
3. If z is real & positive, $\arg(z) = 0$
4. If $\theta = \frac{\pi}{2}$, z lies on the positive side of imaginary axis.
5. If $\theta = -\frac{\pi}{2}$, z lies on the negative side of imaginary axis.

Q.2 For $z = -1 + i\sqrt{3}$

Ans. $\frac{2\pi}{3}$

Sol. $|z| = 2$

$\arg(z) = 2n\pi + \frac{2\pi}{3}, \quad n \in I$

Least positive argument $= \frac{2\pi}{3}$

$\text{amp}(z) = \frac{2\pi}{3}$

Q.3 For $z = 1 - i\sqrt{3}$

Ans. $-\frac{\pi}{3}$

Sol. $|z| = 2$

$\arg(z) = 2n\pi - \frac{\pi}{3}, \; n \in I$

Least positive argument $= \frac{5\pi}{3}$

If the point lies in third or fourth quadrant then consider amp(z) in clockwise direction.

In this case $\text{amp}(z) = -\frac{\pi}{3}$

Q.4 For $z = -1 - i\sqrt{3}$

Ans.

Sol. $|z| = 2$

$\arg(z) = 2n\pi - \frac{2\pi}{3}$, $n \in I$

Least positive argument $= \frac{4\pi}{3}$

$\text{amp}(z) = -\frac{2\pi}{3}$

2.4 POLAR FORM AND EXPONENTIAL FORM OF COMPLEX NUMBER

Polar Form

$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$;

Note : $\cos\theta + i\sin\theta$ is also written as $CiS\,\theta$.

Euler's formula :

The formula $e^{ix} = \cos x + i\sin x$ is called Euler's formula.

Also $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ & $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ are known as Euler's identities.

Exponential Form

Let z be a complex number such that $|z| = r$ & $\arg z = \theta$, then $z = r.e^{i\theta}$ or $z = |z|.e^{i\theta}$

Ex.1 The polar form of $-1 + i$ is-

Ans.

Sol. $\because |-1+i| = \sqrt{2}$, $\text{amp}(-1+i) = \pi - \pi/4 = 3\pi/4$

$\therefore -1 + i = \sqrt{2}\ (\cos 3\pi/4 + i\ \sin 3\pi/4)$ **Ans. [C]**

Properties of Argument :

1. $\arg(z_1.z_2) = \arg z_1 + \arg z_2$
2. $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$
3. $\arg(z^n) = n\arg(z)$; $n \in I$
4. $\arg(\bar{z}) = -\arg(z)$

Properties of Amplitude :

1. $\text{amp}(z_1.z_2) = \text{amp}\ z_1 + \text{amp}\ z_2 + 2k\pi$; $k \in I$
2. $\text{amp}\left(\frac{z_1}{z_2}\right) = \text{amp}\ z_1 - \text{amp}\ z_2 + 2k\pi$; $k \in I$
3. $\text{amp}(z^n) = n\,\text{amp}(z) + 2k\pi$; $n,k \in I$

where proper value of k must be chosen so that RHS lies in $(-\pi, \pi]$.

Ex.1 Convert into Polar form

1. $z = \cos\theta - i\sin\theta \Rightarrow z = \cos(-\theta) + i\sin(-\theta)$

2. $z = \sin\theta + i\cos\theta \Rightarrow \quad z = \cos\left(\frac{\pi}{2} - \theta\right) + i\sin\left(\frac{\pi}{2} - \theta\right)$

3. $z = \sin\theta - i\cos\theta \Rightarrow \quad z = (-i)(\cos\theta + i\sin\theta) \Rightarrow z = e^{-i\frac{\pi}{2}} e^{i\theta} \Rightarrow z = e^{i\left(\theta - \frac{\pi}{2}\right)}$

4. $z = -\cos\theta - i\sin\theta = -1(\cos\theta + i\sin\theta) \Rightarrow z = e^{i\pi} e^{i\theta} = e^{i(\pi + \theta)}$

Ex.2 If $|z| = 1$

(i) $1 + z = (1 + \cos\theta) + i\sin\theta = 2\cos^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$

(ii) $1 - z = 1 - (\cos\theta + i\sin\theta) = (1 - \cos\theta) - i\sin\theta$

$= 2\sin^2\frac{\theta}{2} - 2i\sin\frac{\theta}{2}\cos\theta = 2\sin\frac{\theta}{2}\left(\sin\frac{\theta}{2} - i\cos\frac{\theta}{2}\right)$

Ex.3 $z = -3\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right) \Rightarrow z = 3e^{i\pi} e^{i\frac{5\pi}{6}} \Rightarrow z = 3e^{i\left(\pi + \frac{5\pi}{6}\right)}$

Questions

1. $(x + 2) + (y - 1)i = 5 - 4i, \quad x, y \in R$ Find x, y.

Ans. $x = 3, y = -3$

Sol. Comparing real & Imaginary part

$x + 2 = 5 \quad \& \quad y - 1 = -4$

$x = 3 \quad \& \quad y = -3$

2. Find the least postive $n \in N$ if $\left(\frac{1+i}{1-i}\right)^n = 1$

Ans. $n = 4$

Sol. $\left[\frac{1+i}{1-i}\right]^n = 1$

$\left[\frac{1+i}{1-i} \times \frac{1+i}{1+i}\right]^n = 1$

$\left[\frac{(1+i)^2}{2}\right]^n = 1$

$\left[\frac{1-1+2i}{2}\right]^n = 1$

$i^n = 1$

$n = 4$

3. $z = a + i, a \in R$. If $Re(z) = Re(z^2)$, Find a.

Ans. $a = \frac{1 \pm \sqrt{5}}{2}$

Sol. $z = a + i\ ;\ a \in R$

$Re(z) = a$

Now $z^2 = (a + i)^2$

$= a^2 + i^2 + 2ai$

$= a^2 - 1 + 2ai$

$Re(z^2) = a^2 - 1$

Now, $Re(z) = Re(z^2)$

$a = a^2 - 1$

$a^2 - a - 1 = 0$

$$a = \frac{1 \pm \sqrt{1 - 4(-1)}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

4. If $\frac{(x-2)+(y-3)i}{1+i} = 1 - 3i$, Find (x, y).

Ans : $x = 6\ ;\ y = 1$

Sol. $$\frac{(x-2)+(y-3)i}{1+i} = 1 - 3i$$

$$\frac{\{(x-2)+(y-3)i\}\{1-i\}}{(1+i)(1-i)} = 1 - 3i$$

$$\frac{\{(x-2)+(y-3)\}+i\{(y-3)-(x-2)\}}{2} = 1 - 3i$$

$\therefore x + y - 5 = 2$ & $y - x - 1 = -6$

$x + y = 7$ & $y - x = -5$

$x - y = 5$

Solving $x = 6\ ;\ y = 1$

5 If $\frac{x+iy}{x-iy} = a+ib$, Prove that : $a^2 + b^2 = 1$

Ans.

Sol. $$\frac{x+iy}{x-iy} = a+ib \qquad \text{.....(i)}$$

Take conjvgate on both side

$$\overline{\left(\frac{x+iy}{x-iy}\right)} = \overline{(a+ib)}$$

$$\frac{x-iy}{x+iy} = a - ib \qquad \text{.....(ii)}$$

Multiply (i) and (ii)

$$\frac{(x+iy)(x-iy)}{(x-iy)(x+iy)} = (a+ib)(a-ib)$$

$1 = a^2 + b^2$

6. If $z^2 + 2(1+2i)z - (11+2i) = 0$, find z in the form of $a + ib$.

Ans : $z = 2 - i$ or $z = -(4+3i)$

Sol. $z^2 + 2(1+2i)z - (11+2i) = 0$

$$= \frac{-2(1+2i) \pm \sqrt{4(1+2i)^2 + 4(11+2i)}}{2}$$

$$= -(1+2i) \pm \sqrt{(1+2i)^2 + (11+2i)}$$

$$= -(1+2i) \pm \sqrt{1-4+4i+11+2i}$$

$$= -(1+2i) \pm \sqrt{6i+8}$$

Note:-

$$\sqrt{a+ib} = \pm\left[\sqrt{\frac{|z|+a}{2}} + \frac{ib}{|b|}\sqrt{\frac{|z|-a}{2}}\right]$$

where $z = a + ib$

$\therefore \sqrt{8+6i}$

$$= \pm\left[\sqrt{\frac{10+8}{2}} + i\sqrt{\frac{10-8}{2}}\right]$$

$= \pm(3 + i)$

Now $z = -(1+2i) \pm (3+i)$

$= -1 - 2i + 3 + i$, $\quad -1 - 2i - 3 - i$

$= 2 - i, -4 - 3i$

7. If $f(x) = x^4 - 4x^3 + 4x^2 + 8x + 44$, find $f(3+2i)$.

Ans: 5

Sol. $x = 3 + 2i$

$x - 3 = 2i$

$(x-3)^2 = -4$

$x^2 - 6x + 13 = 0$

Now we have to find

$f(x) = x^4 - 4x^3 + 4x^2 + 8x + 44$

$$
\begin{array}{r}
x^2+2x+3 \\
x^2-6x+13\overline{)\,x^4-4x^3+4x^2+8x+44} \\
x^4-6x^3+13x^2 \\
-\quad +\quad - \\
\hline
2x^3-9x^2+8x+44 \\
2x^3-12x^2+26x \\
-\quad +\quad - \\
\hline
3x^2-18x+44 \\
3x^2-18x+39 \\
-\quad +\quad - \\
\hline
5
\end{array}
$$

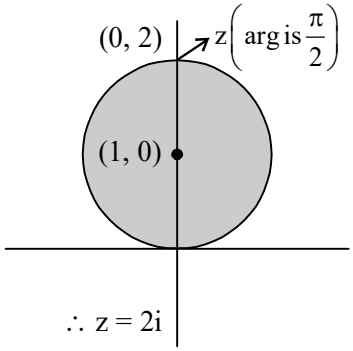
Now, $x^4-4x^3+4x^2+8x+44=\left(x^2-6x+13\right)\left(x^2+2x+3\right)+5$

$\Rightarrow 5n$

8. If $|z-i|=1$ and $\text{Arg } z=\dfrac{\pi}{2}$, find the number of complex number.

Ans. $z=2i$

Sol. $|z-i|=1$ & $\arg z=\dfrac{\pi}{2}$



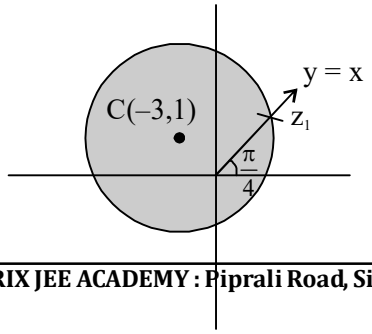
9. If $\text{Arg } z=\dfrac{\pi}{4}$ and $|z+3-i|=4$, find z.

Ans. $z=1+i$

Sol. $\text{Arg } z=\dfrac{\pi}{4}$ & $|z+3-i|=4$

$|z-(-3+i)|=4$

Centre : (–3, 1) & radius = 4



$(x+3)^2+(y-1)^2=4^2$
and straight line $y=x$ (and point of Intersects in first equal)
$(y+3)^2+(y-1)^2=16$
$2y^2+10+4y-16=0$
$y^2+2y-3=0$
$(y+3)(y-1)=0$
$y=1 \quad \therefore x=1$
$\therefore z(1,1)$ or $1+i$

10. (i) $z=\left[\dfrac{(8+i)(7+i)}{(5-i)}\right]^6$ find amp(z) (ii) $z=\dfrac{(1+i)(1+2i)(1+3i)}{(1-i)(2-i)(3-i)}$ find amp(z)

(iii) The value of $\left(\dfrac{1+\sin\dfrac{2\pi}{9}+i\cos\dfrac{2\pi}{9}}{1+\sin\dfrac{2\pi}{9}-i\cos\dfrac{2\pi}{9}}\right)^3$ is :

Ans. (i) $6\tan^{-1}\left(\dfrac{1}{2}\right)$ (ii) $\dfrac{3\pi}{2}$ (iii) $-\dfrac{1}{2}\left(\sqrt{3}-i\right)$

Sol. (i) $z=\left[\dfrac{(8+i)(7+i)}{5-i}\right]^6$

find amp(z)

$$\text{amp}(z)=6\left\{\tan^{-1}\frac{1}{8}+\tan^{-1}\frac{1}{7}-\tan^{-1}\left(-\frac{1}{5}\right)\right\}$$

$$\Rightarrow 6\left\{\tan^{-1}\left(\frac{\frac{1}{8}+\frac{1}{7}}{1-\frac{1}{8.7}}\right)+\tan^{-1}\frac{1}{5}\right\}$$

$$\Rightarrow 6\left\{\tan^{-1}\left(\frac{15}{55}\right)+\tan^{-1}\frac{1}{5}\right\}$$

$$\Rightarrow 6\left\{\tan^{-1}\left\{\frac{\frac{3}{11}+\frac{1}{5}}{1-\frac{3}{55}}\right\}\right\}$$

$$\Rightarrow 6\tan^{-1}\left(\frac{26}{52}\right)$$

$$\Rightarrow 6\tan^{-1}\left(\frac{1}{2}\right)$$

(ii) $z=\left(\frac{1+i}{1-i}\right).\left(\frac{1+2i}{2-i}\right).\left(\frac{1+3i}{3-i}\right)$

$=\left(\frac{1+i}{1-i}\right)\frac{i}{i}\times\left(\frac{1+2i}{2-i}\right)\frac{i}{i}\times\left(\frac{1+3i}{3-i}\right)\frac{i}{i}$

$=\frac{(1+i)(1+2i)(1+3i)i^3}{(1+i)(1+2i)(1+3i)}$

$=i^3$

$=-i$

$\arg z=\frac{3\pi}{2}$

(iii) Let $\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}=z$

$\left(\frac{1+z}{1+\bar{z}}\right)^3=\left(\frac{1+z}{1+\frac{1}{z}}\right)^3=z^3$

$\Rightarrow\left(i\left(\cos\frac{2\pi}{9}-i\sin\frac{2\pi}{9}\right)\right)^3$

$=-i\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)=-i\left(\frac{-1}{2}-i\frac{\sqrt{3}}{2}\right)$

$\Rightarrow\frac{-1}{2}(\sqrt{3}-i)$

11. $z=\frac{(2\sqrt{3}+2i)^8}{(1-i)^6}+\frac{(1+i)^6}{(2\sqrt{3}-2i)^8}$ find amp(z)

Ans. $\frac{5\pi}{6}$

Sol. $z=\frac{\left(2\sqrt{3}+2i\right)^8}{(1-i)^6}+\frac{(1+i)^6}{\left(2\sqrt{3}-2i\right)^8}$

$=\frac{\left\{\left(2\sqrt{3}+2i\right)\left(2\sqrt{3}-2i\right)\right\}^8+\{(1+i)(1-i)\}^6}{(1-i)^6\left(2\sqrt{3}-2i\right)^8}$

$$= \frac{\{16\}^8 \{2\}^6}{(1-i)^6 \left(2\sqrt{3}-2i\right)^8}$$

$$\arg(z) = -6\tan^{-1}(-1) - 8\tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right)$$

$$= 6\tan^{-1}1 + 8\tan^{-1}\frac{1}{\sqrt{3}}$$

$$= 6\cdot\frac{\pi}{4} + 8\cdot\frac{\pi}{6}$$

$$= \frac{3\pi}{2} + \frac{4\pi}{3}$$

$$= \frac{9\pi + 8\pi}{6}$$

$$= \frac{17\pi}{6}$$

$$= 2\pi + \frac{5\pi}{6} \Rightarrow \frac{5\pi}{6}$$

12. If $\arg z = \theta$, $|z| = 1$, $\theta \in \left[0, \frac{\pi}{2}\right]$. Find $\arg\left(\frac{1}{1+z}\right)$

Ans $-\frac{\theta}{2}$

Sol. $\arg z = \theta$; $|z| = 1$

$$\arg\left(\frac{1}{1+z}\right)$$

$$\arg\left(\frac{1}{2\cos\frac{\theta}{2}}.e^{i\frac{\theta}{2}}\right)$$

arg is $-\frac{\theta}{2}$

13. If $z = e^{2+3i}$, find $|z|$ and arg (z).

Ans. $|z| = e^2$; $\arg(z) = 3$

Sol. $z = e^{2+3i}$

$= e^2 \cdot e^{i3}$

$|z| = e^2$ & $\arg(z) = 3$

14. If $i^{i^{i^{\cdots^{\infty}}}} = A + iB$ then Prove that (i) $A^2 + B^2 = e^{-\pi B}$ (ii) $\frac{B}{A} = \tan\left(\frac{\pi A}{2}\right)$

Ans.

Sol. $\left(e^{i\frac{\pi}{2}}\right)^{A+iB} = A + iB$

$e^{i\frac{\pi}{2}A - \frac{\pi B}{2}} = A + iB$

$e^{-\frac{\pi B}{2}}\left[\cos\frac{\pi}{2}A + i\sin\frac{\pi A}{2}\right] = A + iB$

$A = e^{-\frac{\pi B}{2}}\cos\frac{\pi}{2}A; B = +e^{-\frac{\pi B}{2}}\sin\frac{\pi}{2}A$

(i) $A^2 + B^2$

$= e^{-\pi B}\cos^2\frac{\pi A}{2} + e^{-\pi B}\sin^2\frac{\pi A}{2}$

$= e^{-\pi B}\left(\cos^2\frac{\pi A}{2} + \sin^2\frac{\pi A}{2}\right)$

$= e^{-\pi D}$

(ii) $\frac{B}{A} = \frac{e^{-\frac{\pi B}{2}}\sin\pi\frac{A}{2}}{e^{-\frac{\pi B}{2}}\cos\pi\frac{A}{2}}$

$= \tan\frac{\pi A}{2}$

15. $z^3 + iz - 1 = 0$ Find number of real/purely imaginary solutions.

Ans. Number of real root = 0

Numer of purely imaginary root = 0

Sol. $z^3 + iz - 1 = 0$ .....(1)

det real solution of eq.(1) be λ

$z = \lambda$ ratio of eq. (1)

$\lambda^3 + i\lambda - 1 = 0$

$\lambda^3 - 1 + i\lambda = 0$

$\Rightarrow \lambda^3 - 1 = 0$ & $\lambda = 0$

$\lambda^3 = 1$ & $\lambda = 0$

$\lambda = 1$ & $\lambda = 0$

Not possible simultancorly

Again

det purely imaginary solution be

$z = \lambda i$

$(\lambda i)^3 + i(\lambda i) - 1 = 0$

$-i\lambda^3 - \lambda - 1 = 0$

$(\lambda + 1) + i\lambda^3 = 0$

$\Rightarrow \lambda = -1$ & $\lambda = 0$

Not possible simultancorly

$\therefore$ No Real & purely imaginary

16. Using $\ell n\,(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots\ldots$ Prove that : $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \ldots\ldots\infty$

Ans.

Sol. $\ell n\,(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots\ldots$ (i)

$x \to -x$

$\ell n(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots\ldots$ (ii)

(i) – (ii)

$$\ell n\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^3}{5} + \frac{x^7}{7} + \cdots\right)$$

Put $x = i$

$$\ell n\left(\frac{1+i}{1-i}\right) = 2\left(i - \frac{i}{3} + \frac{i}{5} - \frac{i}{7} + \cdots\right)$$

$$\ell n\left(\frac{(1+i)^2}{2}\right) = 2\left(i - \frac{i}{3} + \frac{i}{5} - \frac{i}{7} + \cdots\right)$$

$$\ell n(i) = 2i\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right]$$

$$\ell n\left[e^{i\pi/2}\right] = 2i\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right]$$

$$i\frac{\pi}{2} = 2i\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right]$$

$$\frac{\pi}{4} = 1 - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Proved

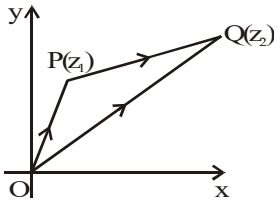
3. GEOMETRICAL INTERPRETATION

If $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$

$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

$|z_1 - z_2|$ denotes distance between points z_1 and z_2 on complex plane.

If $P(z_1)$ & $Q(z_2)$ be two complex numbers on argand plane then



$\vec{PQ}$ represents complex number $z_2 - z_1$.

Ex.1 If $|z| = 2$, find locus of z.

Ans. $x^2 + y^2 = 4$

Sol. det $z = x + iy$

$|z| = 2$

$|x + iy| = 2$

$\sqrt{x^2 + y^2} = 2$

$x^2 + y^2 = 4$

Ex.2 If $|z - z_1| = r$, find locus of z.

Ans. Circle whose center is z_1, & radius is r

Sol. Centre : z_1

Radius : r

Ex.3 If $|2z - 1| = 3$, find locus of z.

Ans. circle whose center is $\left(\frac{1}{2}, 0\right)$ and radius is 3/2

Sol. $|2z - 1| = 3$

$2\left|z - \frac{1}{2}\right| = 3$

$\left|z - \frac{1}{2}\right| = \frac{3}{2}$

Centre : $\left(\frac{1}{2}, 0\right)$

Realine : $\frac{3}{2}$

Note: $A(z_1)$ and $B(z_2)$ are fixed points, P(z) is a variable pt.

1. $\frac{PA}{PB} = K$ or $\left|\frac{z - z_1}{z - z_2}\right| = K$.

(i) $K = 1 \Rightarrow$ locus of P is a line.

(ii) $K \neq 1 \Rightarrow$ locus of P is a circle.

2. $PA + PB = K$ or $|z - z_1| + |z - z_2| = K$

(i) $K > AB \Rightarrow$ locus of P is an ellipse

(ii) $K = AB \Rightarrow$ locus of P is line segment AB

(iii) $K < AB \Rightarrow$ No locus

3. $|PA - PB| = K$ or $\big||z - z_1| - |z - z_2|\big| = K$

(i) $K < AB \Rightarrow$ locus of P is hyperbola.

(ii) $K = AB \Rightarrow$ Locus of P will be line AB excluding all points between A and B

(iii) $K > AB \Rightarrow$ No locus

Section Formulae

If z_1 and z_2 are affixes of the two points P and Q respectively and

1. Point C divides the line segment joining P and Q internally in the ratio m : n then affix z of C is given by $z = \frac{mz_2 + nz_1}{m+n}$ where $m, n > 0$

2. If C divides PQ in the ratio m : n externally then $z = \frac{mz_2 - nz_1}{m-n}$

Triangle

If the vertices A, B, C of a Δ are represented by complex numbers z_1, z_2, z_3 respectively and a, b, c are the length of sides then,

(i) Centroid $'G' = \frac{z_1 + z_2 + z_3}{3}$

(ii) Incentre $'I' = \frac{a z_1 + b z_2 + c z_3}{a + b + c}$

(iii) Orthocentre :

$$Z_H = \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\Sigma \tan A} \quad \text{or} \quad \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\Pi \tan A}$$

(iv). Circumcentre

Hence $Z_D = \frac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\Sigma \sin 2A}$

Triangle Inequality

If $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

In triangle OAC

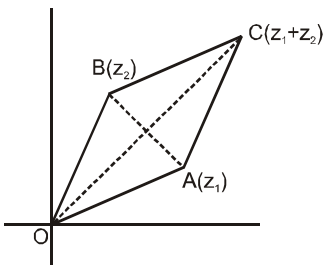
$OC \le OA + AC$

$OA \le AC + OC$

$AC \le OA + OC$

using these in equalities we have

$||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$



Note : (a) $||z_1| - |z_2|| = |z_1 + z_2|$, iff origin, z_1 and z_2 are collinear and origin lies between z_1 and z_2.

(b) $|z_1 + z_2| = |z_1| + |z_2|$, iff origin, z_1 and z_2 are collinear and z_1 and z_2 lies on the same side of origin.

Questions

Q.1 Find locus of P(z) if $\text{Re}(z+i) = |z-1|$

Ans. $y^2 = 2x - 1$

Sol. $\det z = x + iy$

$\text{Re}(z+i) = |z-1|$

$\text{Re}(x + iy + i) = |x + iy - 1|$

$\text{Re}(x + i(y+1)) = |x - 1 + iy|$

$x = \sqrt{(x-1)^2 + y^2}$

$x^2 = (x-1)^2 + y^2$

$x^2 = x^2 - 2x + 1 + y^2$

$y^2 = 2x - 1$

Q.2. Find locus of P(z) if $|z-3| = |z-2+i|$

Ans. $x + y = 2$

Sol. $|z-3| = |z-2+i|$

$\det z = x + iy$

$|x + iy - 3| = |x + iy - 2 + i|$

$|x - 3 + iy| = |x - 2 + i(y+1)|$

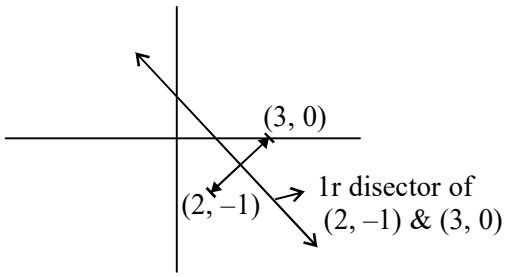
$(x-3)^2 + y^2 = (x-2)^2 + (y+1)^2$

$x^2 - 6x + 9 + y^2 = x^2 - 4x + 4 + y^2 + 2y + 1$

$2x + 2y = 4$

$x + y = 2$

OR



Q.3 Find locus of P(z) if $\left|\frac{z-2}{z-1}\right| = 2$

Ans. $3x^2 + 3y^2 = 4x$

Sol. $\left|\frac{z-2}{z-1}\right| = 2$

$|z-2| = 2|z-1|$

$\det z = x + iy$

$|x + iy - 2| = 2|x + iy - 1|$

$\sqrt{(x-2)^2+y^2} = 2\sqrt{(x-1)^2+y^2}$

$x^2-4x+4+y^2 = 4\{x^2-2x+1+y^2\}$

$3x^2+3y^2-4x = 0$

Q.4 Find locus of P(z) if $|z-3+i|=1$

Ans. $(x-3)^2+(y+1)^2=1$

Sol. $|z-3+i|=1$

Circle with centre (3, –1) & Rationl

$(x-3)^2+(y+1)^2=1$

Q.5 Find locus of P(z) if $\text{Re}\left(\frac{1}{z}\right)=\frac{1}{2}$

Ans. $x^2+y^2-2x=0$

Sol. $\text{Re}\left(\frac{1}{z}\right)=\frac{1}{2}$

Let $z = x+iy$

$\text{Re}\left(\frac{1}{x+iy}\cdot\frac{x-iy}{x-iy}\right)=\frac{1}{2}$

$\text{Re}\left(\frac{x-iy}{x^2+y^2}\right)=\frac{1}{2}$

$\frac{x}{x^2+y^2}=\frac{1}{2}$

$x^2+y^2-2x=0$

Q.6 Find z which satisfy simultaneously $\left|\frac{z-12}{z-8i}\right|=\frac{5}{3}$ and $\left|\frac{z-4}{z-8}\right|=1$

Ans. $6+8i,\quad 6+17i$

Sol. $3|z-12| = 5|z-8i|$ & $|z-4| = |z-8|$

$3|(x-12)+iy| = 5|x+i(y-8)|$

$9\{(x-12)^2 y^2\} = 25\ \{x^2+(y-8)^2\}$

$9\{x^2-24x+144+y^2\} = 25\{x^2+y^2-16y+64\}$

$16x^2+16y^2+216x-400y+304=0$

$2x^2+2y^2+27x-50y+38=0$ .....(A)

& $|z-4| = |z-8|$

$|x+iy-4| = |x+iy-8|$

$(x-4)^2+y^2 = (x-8)^2+y^2$

$x-4 = 8-x$

$2x = 12$

$x = 6$ .....(B)

Put in eq. (B)

$2.6^2 + 2y^2 + 27.6 - 50y + 38 = 0$

$2y^2 - 50y + 272 = 0$

$y^2 - 25y + 136 = 0$

$y^2 - 8y - 17y + 136 = 0$

$y(y - 8) - 17(y - 8) = 0$

$y = 8, 17$

$\therefore z = 6 + 8i \ \& \ 6 + 17i$

Q.7 $|z - 3| = |z - 2 + i|$, find min value of $|z|$

Ans. $\sqrt{2}$

Sol. $|z - 3| = |z - (2 - i)|$

Let $z = x + iy$

$|x + iy - 3| = |(x - 2) + i(y + 1)|$

$(x - 3)^2 + y^2 = (x - 2)^2 + (y + 1)^2$

$x^2 - 6x + 9 + y^2 = x^2 - 4x + 4 + y^2 + 2y + 1$

$2x + 2y - 4 = 0$

$x + y = 2 \qquad \Rightarrow x + y - 2 = 0$

Now $|z| = \dfrac{|0 + 0 - 2|}{\sqrt{1 + 1}}$

$= \dfrac{2}{\sqrt{2}}$

$= \sqrt{2}$

Q.8 $z + |z| = 8 + 4i$; find z.

Ans. $z = 3 + 4i$

Sol. $z + |z| = 8 + 4i$

$x + iy + \sqrt{x^2 + y^2} = 8 + 4i$

$x + \sqrt{x^2 + y^2} = 8 \ \& \ y = 4$

$\sqrt{x^2 + 16} = 8 - x$

$x^2 + 16 = 64 - 16x + x^2$

$16x = 48$

$x = 3 \ \& \ y = 4$

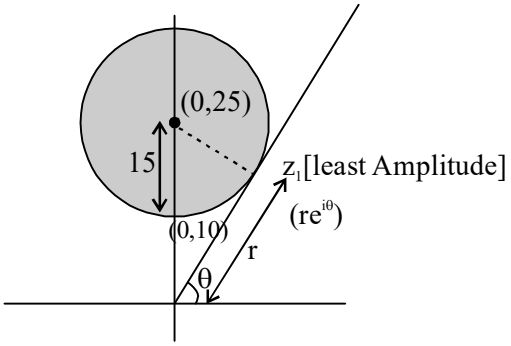
$\therefore z = 3 + 4i$

Q.9 Among the complex number z which satisfy $|z - 25i| \le 15$, find the one

(i) having the least amplitude (ii) having greatest amplitude (iii) having greatest modulus

Ans. (i) $12 + 16i$ (ii) $-12 + 16i$ (iii) $40i$

Sol. **(i)** $\sin(90^\circ - \theta) = \frac{15}{25}$



$\cos\theta = \frac{3}{5}$

$\sin\theta = \frac{4}{5}$ & $25^2 = r^2 + 15^2$

$= \sqrt{625 - 225}$

$= 20$

$\therefore Z_1 = re^{i\theta}$

$= 20\left[\frac{3}{5} + i\frac{4}{5}\right]$

$= 4(3 + i4)$

$= 12 + i16$

(ii) complex number with greatest amplitude will be mirror image of z_1 wrt y–axis

$\therefore -12 + 16i$

(iii) complex number with greatest modulus

$= 25i + 15i$

$= 40i$

Q.10 Show that the locus of the point P(ω) denoting the complex number $z + \frac{1}{z}$ on the complex plane is a standard ellipse where $|z| = a$, where $a \neq 0$, 1

Ans. 1

Sol. Let $w = z + \frac{1}{z}$ where $z = \alpha + i\beta$, $\alpha^2 + \beta^2 = a^2$ (as $|z| = a$)

$$x + iy = \alpha + i\beta + \frac{1}{\alpha + i\beta} = \alpha + i\beta + \frac{\alpha - i\beta}{\alpha^2 + \beta^2} = \left(\alpha + \frac{\alpha}{a^2}\right) + i\left(\beta - \frac{\beta}{a^2}\right)$$

$$\therefore \quad x = \alpha\left(1 + \frac{1}{a^2}\right) \quad ; \quad y = \beta\left(1 - \frac{1}{a^2}\right)$$

$$\therefore \quad \frac{x^2}{\left(1 + \frac{1}{a^2}\right)^2} + \frac{y^2}{\left(1 - \frac{1}{a^2}\right)^2} = \alpha^2 + \beta^2 = a^2$$

$$\therefore \quad \frac{y^2}{\left(a + \frac{1}{a}\right)^2} + \frac{y^2}{\left(a - \frac{1}{a}\right)^2} = 1$$

Q.11 PT : $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\,(|z_1|^2 + |z_2|^2)$

Ans. $2\left(|z_1|^2 + |z_2|^2\right)$

Sol. $|z_1 + z_2|^2 + |z_1 - z_2|^2$

$\Rightarrow (z_1+z_2)\left(\overline{z_1+z_2}\right)+(z_1-z_2)\left(\overline{z_1-z_2}\right)$

$\Rightarrow (z_1+z_2)(\overline{z}_1+\overline{z}_2)+(z_1-z_2)(\overline{z}_1-\overline{z}_2)$

$\Rightarrow z_1\overline{z}_1+z_1\overline{z}_2+z_2\overline{z}_1+z_2\overline{z}_2+z_1\overline{z}_1-z_1\overline{z}_2-\overline{z}_1z_2+z_2\overline{z}_2$

$\Rightarrow 2\left(|z_1|^2+|z_2|^2\right)$

Q.12 If $\left|\dfrac{z_1-z_2}{1-\overline{z}_1 z_2}\right|<1$ then prove that $|z_1|<1$ and $|z_2|<1$ **or** $|z_1|>1$ and $|z_2|>1$

Ans. $|z_1|<1 \quad \& \quad |z_2|<1$

Sol. $\left|\dfrac{z_1-z_2}{1-\overline{z}_1z_2}\right|<1$

$|z_1-z_2|<|1-\overline{z}_1z_2|$

$(z_1-z_2)\overline{(z_1-z_2)}<\left(1-\overline{z}_1z_2\right)\overline{\left(1-\overline{z}_1z_2\right)}$

$(z_1-z_2)\left(\overline{z_1}-\overline{z_2}\right)<\left(1-\overline{z}_1z_2\right)\left(1-z_1\overline{z_2}\right)$

$z_1\overline{z}_1-z_1\overline{z}_2-z_2\overline{z}_1+z_2\overline{z}_1<1-z_1\overline{z}_2-\overline{z}_1z_2+\overline{z}_1z_1z_2\overline{z}_2$

$|z_1|^2+|z_2|^2<1+|z_1|^2|z_2|^2$

$\left(|z_1|^2-1\right)+|z_2|^2\left(1-|z_1|^2\right)<0$

$\left(|z_1|^2-1\right)\left(1-|z_2|^2\right)<0$

$\left(|z_1|^2-1\right)\left(|z_2|^2-1\right)>0$

C–1

$\Rightarrow |z_1|^2-1>0 \,\&\, |z_2|^2-1>0$

$|z_1|>1 \quad \& \quad |z_2|>1$

Or

C–2

$|z_1|^2-1<0 \quad \& \quad |z_2|^2-1<0$

$|z_1|<1 \quad \& \quad |z_2|<1$

Q.13 If $\left|\dfrac{6z-i}{2+3iz}\right|\le 1$ then prove that $|z|\le\dfrac{1}{3}$.

Ans.

Sol. $|6z-i|\le|2+3iz| \quad \Rightarrow \quad |6z-i|^2\le|2+3iz|^2$

$(6z-i)(6\overline{z}+i)\le(2+3iz)((2-3i\overline{z})$

$$36z\bar{z} + 6zi - 6i\bar{z} + 1 \le 4 - 6i\bar{z} + 6iz - 9i^2 z\bar{z}$$

$$36z\bar{z} + 1 \le 4 + 9iz\bar{z}$$

$$27z\bar{z} \le 3$$

$$|z|^2 \le 1/9$$

$$|z| \le 1/3$$

Q.14 Let z be a complex number such that $z \in C \setminus R$ and $\dfrac{1+z+z^2}{1-z+z^2} \in R$ then prove that $|z| = 1$.

Ans.

Sol We have $\dfrac{1+z+z^2}{1-z+z^2} = \dfrac{1-z+z^2+2z}{1-z+z^2} = 1 + \dfrac{2}{z-1+(1/z)}$

Hence $z + \dfrac{1}{z} \in R$; $\quad\therefore\quad z + \dfrac{1}{z} = \bar{z} + \dfrac{1}{\bar{z}}$

$$(z - \bar{z}) = \frac{1}{\bar{z}} - \frac{1}{z} = \frac{z - \bar{z}}{|z|^2} \quad \Rightarrow \quad (z - \bar{z})\,(|z|^2 - 1) = 0$$

$\therefore \quad |z| = 1; \quad \therefore \quad z = \bar{z}$ (not possible as z is not real)]

Q.15 Let $z_1, z_2, z_{3,\ldots..} z_n$ are the complex numbers such that $|z_1| = |z_2| = \ldots.. = |z_n| = 1$. Then

(i) Prove that $\dfrac{z_1 + z_2}{1 + z_1 z_2}$ is real

(ii) Prove that $\left|z_1 + z_2 + \ldots\ldots..z_n\right| = \left|\dfrac{1}{z_1} + \dfrac{1}{z_2} + \ldots\ldots..\dfrac{1}{z_n}\right|$

(iii) Prove that $\left(\sum_{k=1}^{n} z_k\right)\left(\sum_{k=1}^{n} \dfrac{1}{z_k}\right)$ is real

Ans.

Sol (i) $\det\ w = \dfrac{z_1 + z_2}{1 + z_1 z_2}$

$$\therefore \bar{w} = \frac{\bar{z}_1 + \bar{z}_2}{1 + \bar{z}_1 \bar{z}_2}$$

Now $|z_1| = 1$

$\therefore z_1 \bar{z}_1 = 1$

$$\bar{z}_1 = \frac{1}{z_1} \,\&\, \bar{z}_2 = \frac{1}{z_2}$$

$\therefore w - \bar{w}$

$$= \frac{z_1 + z_2}{1 + z_1 z_2} - \frac{\frac{1}{z_1} + \frac{1}{z_2}}{1 + \frac{1}{z_1} \cdot \frac{1}{z_2}}$$

$$= \frac{z_1 + z_2}{1 + z_1 z_2} - \frac{z_1 + z_2}{z_1 z_2 + 1}$$

$= 0$

$\therefore w = \bar{w} \Rightarrow$ Tuely Real

(ii) $\left|z_1 + z_2 + \ldots + z_n\right| = \overline{\left|z_1 + z_2 + \ldots + z_n\right|}$

$$= \left|\bar{z}_1 + \bar{z}_2 + \ldots + \bar{z}_n\right|$$

$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots + \frac{1}{z_n}\right|$$

(iii) $z\bar{z} = 1 \Rightarrow z = \frac{1}{\bar{z}}$

$$\text{now} \quad z = (z_1 + z_2 + z_3 + ... + z_n)\left(\frac{1}{z_1} + \frac{1}{z_2} + ... + \frac{1}{z_n}\right)$$

$$= (z_1 + z_2 + ... + z_n)\left(\bar{z}_1 + \bar{z}_2 + ... + \bar{z}_3\right)$$

$$= (z_1 + z_2 + ... + z_n)\left(\overline{z_1 + z_2 + ... + z_n}\right)$$

$= |z_1 + z_2 + ... + z_n|^2$ which is real

Q.16 If $|z| = 2$, find locus of $w = z + \frac{1}{z}$

Ans. $\frac{4x^2}{25} + \frac{4y^2}{9} = 1$

Sol. If $|z| = 2$

Let $z = 2e^{i\theta}$

Now $w = z + \frac{1}{z}$

$$x + iy = 2e^{i\theta} + \frac{1}{2e^{i\theta}}$$

$$= 2e^{i\theta} + \frac{1}{2}e^{-i\theta}$$

$$= 2(\cos\theta + i\sin\theta) + \frac{1}{2}(\cos\theta - i\sin\theta)$$

$$x = \left(2+\frac{1}{2}\right)\cos\theta \text{ and } y = \left(2-\frac{1}{2}\right)\sin\theta$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\frac{x^2}{\left(\frac{5}{2}\right)^2} + \frac{y^2}{\left(\frac{3}{2}\right)^2} = 1$$

Q.17 For all complex numbers z with $|z| = 1$, Prove that $|1 - z| + |1 + z^2| \le 4$.

Ans.

Sol. Let $z = \cos\theta + i\sin\theta$

$$|1 - z| = |(1 - \cos\theta) - i\sin\theta| = \sqrt{(1-\cos\theta)^2 + \sin^2\theta} = \sqrt{2 - 2\cos\theta} = 2\left|\sin\frac{\theta}{2}\right| \quad ....(1)$$

$$|1 + z^2| = |1 + \cos 2\theta + i\sin 2\theta| = \sqrt{(1+\cos 2\theta)^2 + \sin^2 2\theta} = \sqrt{2 + 2\cos 2\theta} = 2|\cos\theta|$$

$$|1 + z^2| = 2\left|1 - 2\sin^2\frac{\theta}{2}\right| \quad ....(2)$$

from (1) and (2)

$$2\left|\sin\frac{\theta}{2}\right| + 2\left|1 - 2\sin^2\frac{\theta}{2}\right| \le 4$$

Q.18 If $|z| \le 1$; $|w| \le 1$ prove that $|z - w|^2 \le (|z| - |w|)^2 + (\arg z - \arg w)^2$

Ans.

Sol. $z = r(\cos\theta + i\sin\theta) \quad 0 < r \le 1$

$w = R(\cos\phi + i\sin\phi) \quad 0 < R \le 1$

L.H.S. $|(r\cos\theta - R\cos\phi) + i(r\sin\theta - R\sin\phi)|^2$

$= r^2 + R^2 - 2rR\cos(\theta - \phi)$

$$= r^2 + R^2 - 2rR\left[1 - 2\sin^2\left(\frac{\theta-\phi}{2}\right)\right] = (r - R)^2 + 4rR.\sin^2\left(\frac{\theta-\phi}{2}\right)$$

$$\le (|z| - |w|)^2 + 4\left(\frac{\theta-\phi}{2}\right)^2 = (|z| - |w|)^2 + (\arg z - \arg w)^2$$

Q.19 Find the greatest and least values of $|z|$ if z satisfies $\left|z - \frac{4}{z}\right| = 2$.

Ans.

Sol We have $|z| = \left|z - \frac{4}{z} + \frac{4}{z}\right| \le \left|z - \frac{4}{z}\right| + \frac{4}{|z|} = 2 + \frac{4}{|z|}$

$$\Rightarrow \quad |z|^2 \le 2|z| + 4 \Rightarrow \quad (|z| - 1)^2 \le 5$$

$(|z| - 1)^2 \le 5$

$||z| - 1| \le \sqrt{5}$

$$-\sqrt{5} \le 1-|z| \le \sqrt{5}$$

$$|z| \le 1+\sqrt{5} \quad \& \quad |z| \ge 1-\sqrt{5}$$

$$\Rightarrow |z|_{\text{Geatest Vale}} = 1+\sqrt{5} \ \& |z|_{\text{least Value}} = 1-\sqrt{5}$$

Q.20 Let the lines $(2-i)z = (2+i)\ \bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C. If the line $iz + \bar{z} + 1 + i = 0$ is tangent to this circle C, then its radius is :
माना रेखाऐं $(2-i)z = (2+i)\ \bar{z}$ तथा $(2+i)z + (i-2)\bar{z} - 4i = 0$, (यहाँ $i^2 = -1$) एक वृत्त C पर अभिलम्ब हैं। यदि रेखा $iz + \bar{z} + 1 + i = 0$, वृत्त C की स्पर्श रेखा है, तो इसकी त्रिज्या है :

Ans. $\dfrac{3}{2\sqrt{2}}$

Sol. $z = x + iy$

$\bar{z} = x - iy$

$(2-i)z = (2+i)\bar{z}$

$(2-i)(x+iy) = (2+i)(x-iy)$

$x - 2y = 0$ _____(1)

$(2+i)z + (i-2)\bar{z} - 4i = 0$

$(2+i)(x+iy) + (i-2)(x-iy) - 4i = 0$

$x + 2y - 2 = 0$ _____(2)

Point of intersection of (1) and (2) is centre of circle which is $\left(1, \dfrac{1}{2}\right)$

Now, tangent

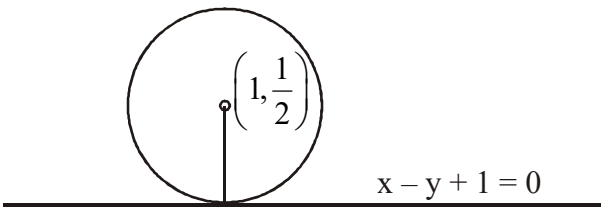
$iz + \bar{z} + 1 + i = 0$

$i(x+iy) + x - iy + 1 + i = 0$

$(x - y + 1) + i(x - y + 1) = 0$

$x - y + 1 = 0$

$$r = \left| \frac{1 - \frac{1}{2} + 1}{\sqrt{2}} \right|$$



$$r = \frac{3}{2\sqrt{2}}$$

Q.21 Let z be those complex numbers which satisfy $|z+5| \le 4$ and $z(1+i) + \bar{z}(1-i) \ge -10$, $i = \sqrt{-1}$. If the maximum value of $|z+1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

माना z वह सभी सम्मिश्र संख्याऐं हैं, जो $|z+5| \le 4$ तथा $z(1+i) + \bar{z}(1-i) \ge -10$, $i = \sqrt{-1}$ को संतुष्ट करती हैं। यदि $|z+1|^2$ का अधिकतम मान $\alpha + \beta\sqrt{2}$ है, तो $(\alpha + \beta)$ का मान है _____।

Ans. 48

Sol. $|z+5| \le 4$

$(x+5)^2 + y^2 \le 16$ .....(i)

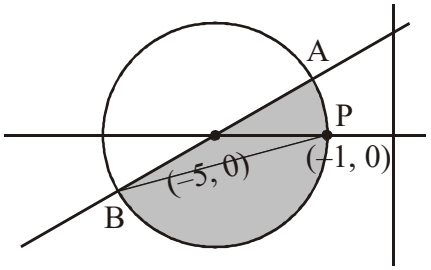
$z(1+i) + \bar{z}(1-i) \ge -10$

$(z + \bar{z}) + i(z - \bar{z}) \ge -10$

$\Rightarrow 2x + i(2iy) \ge -10$

$x - y \ge -5$ .....(ii)

$|z+1|$ = Distance of z from $(-1, 0)$



$|z+1|_{max} = PB$

for B

solve $(x+5)^2 + y^2 = 16$ and $x - y = -5$

$x + 5 = y$

$y^2 + y^2 = 16$

$\therefore\ y = \pm 2\sqrt{2}$

for B, $y = -2\sqrt{2}$

$\left(-5-2\sqrt{2}, -2\sqrt{2}\right)$

$PB = \sqrt{\left(-4-2\sqrt{2}\right)^2 + 8}$

$|z+1|^2_{max} = PB^2 = 8 + \left(4+2\sqrt{2}\right)^2$

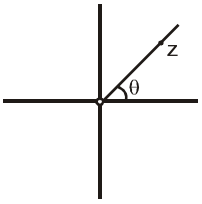
$= 32 + 16\sqrt{2}$

$\alpha = 32,\ \beta = 16$

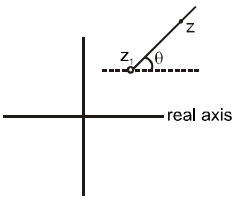
$\therefore\ \alpha + \beta = 48$

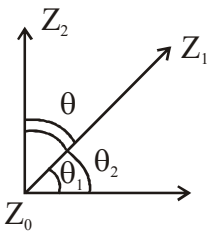
4.1 ANGLES

arg z = θ represents points (non-zero) on ray eminating from origin making an angle θ with positive direction of real axis



$\arg(z - z_1) = \theta$ represents points ($\neq z_1$) on ray eminating from z_1 making an angle θ with positive direction of real axis





Angle between two lines

$\theta_2 = \arg(Z_2 - Z_0)$

$\theta_1 = \arg(Z_1 - Z_0)$

$\theta_2 - \theta_1 = \arg(Z_2 - Z_0) - \arg(Z_1 - Z_0)$

$\arg\left(\frac{Z_2 - Z_0}{Z_1 - Z_0}\right) = \theta \Rightarrow$ $\arg\left(\frac{\text{F.V.}}{\text{I.V.}}\right) = \theta$

Note:

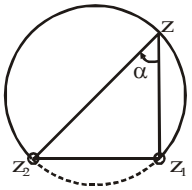
1. If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then

(i) AB $||$ CD if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely real; (ii) AB $\perp$ CD if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary.

2. Note that the locus of z satisfying $\arg\left(\frac{z - z_1}{z - z_2}\right) = \alpha$ is:

Case (a) $0 < \alpha < \pi/2$

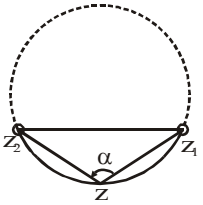
Locus is major arc of circle as shown excluding z_1 & z_2



Case (b) $\frac{\pi}{2} < \alpha < \pi$

Locus is minor arc of circle as shown

excluding z_1 & z_2



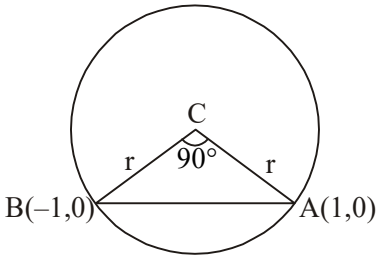
Q.1 The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

समीकरण $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ एक वृत्त को निरूपित करता है जिसका :

Ans. centre at (0, 1) and radius $\sqrt{2}$

Sol. $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

Chord AB will subtend 90° at center.



$2r^2 = (AB)^2 \Rightarrow \quad r = \sqrt{2}$

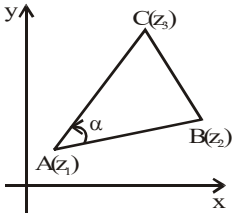
Center = (0, 1)

4.2 ROTATION

If z_1, z_2, z_3 be the three points A,B and C as shown in figure

$$\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \alpha$$

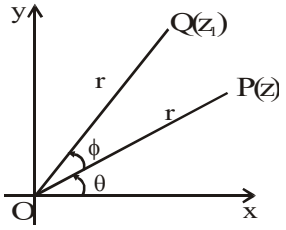
$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\alpha}.$$



$$\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{i\alpha}$$

$$\frac{\text{F.V.}}{|\text{F.V.}|} = \frac{\text{I.V.}}{|\text{I.V.}|} e^{i\alpha}$$

If $\vec{OP} = z = re^{i\theta}$ then $\vec{OQ} = z_1 = re^{i(\theta+\phi)} = z.\, e^{i\phi}$. If $\vec{OP}$ and $\vec{OQ}$ are of unequal magnitude then $\hat{OQ} = \hat{OP}\, e^{i\phi}$ i.e. $\frac{z_1}{|z_1|} = \frac{z}{|z|} e^{i\phi}$



Questions

Q.1 If area of triangle bounded by z, iz, z + iz is 8. Find |z|.

Ans. 4

Q.2 If z_1, z_2, z_3 are the vertices of an equilateral triangle and if z_0 is its circumcentre then

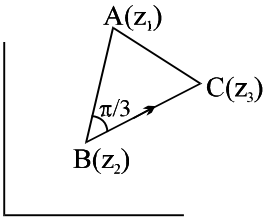
(i) Prove that $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$

(ii) Prove that $3z_0^2 = z_1^2 + z_2^2 + z_3^2$

(iii) Prove that $\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0$

Ans.

Sol. (i) and (ii) $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$



$$\frac{z_1 - z_2}{z_2 - z_3} = \frac{(z_3 - z_2)\, e^{i\pi/3}}{(z_1 - z_3)\, e^{i\pi/3}}$$

$$(z_1 - z_2)(z_1 - z_3) = -(z_2 - z_3)^2$$

$$z_1^2 - z_1 z_3 - z_2 z_1 + z_2 z_3 + z_2^2 + z_3^2 - 2z_2 z_3 = 0$$

$\therefore \quad \sum z_1^2 = \sum z_1 z_2$

Now if z_0 is the circumcentre of the Δ then to prove that $3z_0^2 = z_1^2 + z_2^2 + z_3^2$

Given $z_1 + z_2 + z_3 = 3\, z_0$; $\sum z_1^2 + 2\sum z_1 z_2 = 9z_0^2$; $3\sum z_1^2 = 9z_0^2$

(iii) Prove that $\dfrac{1}{z_2 - z_3} + \dfrac{1}{z_3 - z_1} + \dfrac{1}{z_1 - z_2} = 0$

LHS:- $\dfrac{\overline{z_2 - z_3}}{(z_2 - z_3)(\overline{z_2 - z_3})} + \dfrac{\overline{z_3 - z_1}}{(z_3 - z_1)(z_3 - z_1)} \dfrac{\overline{(z_1 - z_2)}}{(z_1 - z_2)(\overline{z_1 - z_2})}$

$\dfrac{\overline{z}_2 - \overline{z}_3}{|z_2 - z_3|^2} + \dfrac{\overline{z}_3 - \overline{z}_1}{|z_3 - z_1|^2} + \dfrac{\overline{z}_1 - \overline{z}_2}{|z_1 - z_2|^2}$

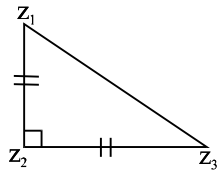
In case of equileterdt triangle

$|z_2 - z_3| = |z_3 - z_1| = |z_1 - z_2| = \lambda$

$\dfrac{1}{\lambda^2}\{\overline{z}_2 - \overline{z}_3 + \overline{z}_3 - \overline{z}_1 + \overline{z}_1 - \overline{z}_2\} \Rightarrow 0$

Q.3 If z_1 , z_2 , z_3 are the vertices of an isosceles triangle right angled at z_2 then prove that $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$

Ans.



Sol $(z_1 - z_2) = (z_3 - z_2)\, e^{i\pi/2}$

$(z_3 - z_2)^2 = i^2 (z_1 - z_2)^2$

$z_3^2 + z_2^2 - 2z_3 z_2 = -z_1^2 - z_2^2 - 2z_1 z_2 \Rightarrow$ result]

Q.4 If z_r(r = 1, 2, ...6) are the vertices of a regular hexagon then prove that $\sum_{r=1}^{6} z_r^2 = 6z_0^2$ where z_0 is the circumcentre.

Ans.

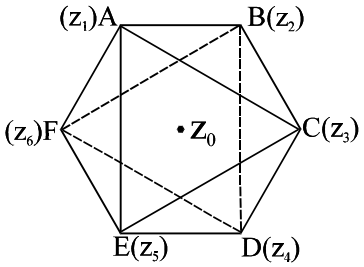
Sol. If z_r(r = 1, 2, ...6) are the vertices of a regular hexagon then

$\sum_{r=1}^{6} z_r^2 = 6z_0^2$ where z_0 is the circumcentre of the regular hexagon.

$3z_0^2 = z_1^2 + z_3^2 + z_5^2$

$3z_0^2 = z_2^2 + z_4^2 + z_6^2$

$6z_0^2 = \sum z_r^2$



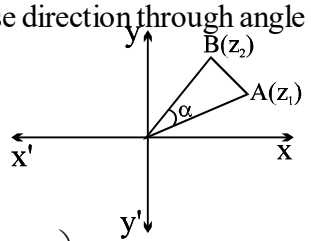
Q.5 Let z_1 and z_2 be roots of the equation $z^2 + pz + q = 0$, where the coefficients p and q may be complex numbers. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and OA = OB, where O is the origin, prove that $p^2 = 4q\cos^2\alpha/2$.

Ans.

Sol. Since z_1 and z_2 are roots of the equation $z^2 + pz + q = 0$

$\therefore \quad z_1 + z_2 = -p$ and $z_1 z_2 = q$ ....(i)

Since OA = OB. So $\vec{OB}$ is obtained by rotating $\vec{OA}$ in anticlockwise direction through angle α.



$\therefore \quad \vec{OB} = \vec{OA}\, e^{i\alpha} \quad \Rightarrow \quad z_2 = z_1 e^{i\alpha}$

$\Rightarrow \frac{z_2}{z_1} = e^{i\alpha} \quad \Rightarrow \quad \frac{z_2}{z_1} = \cos\alpha + i\sin\alpha$

$\Rightarrow \frac{z_2}{z_1} + 1 = 1 + \cos\alpha + i\sin\alpha \Rightarrow \frac{z_2 + z_1}{z_1} = 2\cos\frac{\alpha}{2}\left(\cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2}\right) = 2\cos\frac{\alpha}{2}\, e^{i\alpha/2}$

$\Rightarrow \frac{z_2 + z_1}{z_1} = 2\cos\frac{\alpha}{2}\, e^{i\alpha/2} \Rightarrow \left(\frac{z_2 + z_1}{z_1}\right)^2 = 4\cos^2\frac{\alpha}{2}\, e^{i\alpha}$

$\Rightarrow \left(\frac{z_2 + z_1}{z_1}\right)^2 = 4\cos^2\frac{\alpha}{2}\,\frac{z_2}{z_1} \Rightarrow (z_2 + z_1)^2 = 4 z_1 z_2 \cos^2\frac{\alpha}{2}$

$\Rightarrow \quad (-p)^2 = 4\, q\, \cos^2\frac{\alpha}{2} \quad \Rightarrow \quad p^2 = 4\, q\, \cos^2\frac{\alpha}{2}$

Q.6 A, B, C are the points representing the complex numbers z_1, z_2, z_3 respectively and the circumcentre of the triangle ABC lies at the origin. If the altitudes of the triangle through the opposite vertices meets the circumcircle at D, E, F respectively. Find the complex numbers corresponding to the points D, E, F in terms of z_1, z_2, z_3.

Ans.

Sol $\vec{OD} = \vec{OB}\, e^{i(\pi - 2B)}$

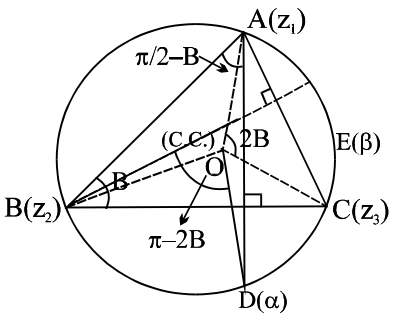
$\alpha = z_2\, e^{i(\pi - 2B)} = -z_2\, e^{-i2B}$

$\alpha = -z_2\, e^{-i2B}$.....(1)

Also, $z_1 = z_3\, e^{i2B}$ .....(2)

$\alpha z_1 = -z_2 z_3 \quad \Rightarrow \quad \alpha = -\frac{z_2 z_3}{z_1}$

$|||^{ly} \quad \beta = -\frac{z_3 z_1}{z_2}$ and $\gamma = -\frac{z_1 z_2}{z_3}$



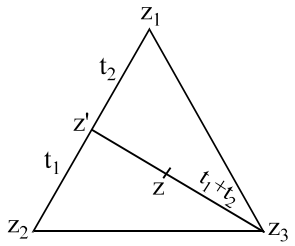
Q.7 Prove that the roots of the equation $\frac{1}{z - z_1} + \frac{1}{z - z_2} + \frac{1}{z - z_3} = 0$ (where z_1, z_2, z_3 are pairwise distinct complex numbers) correspond to points on a complex plane, which lie inside a triangle with vertices z_1, z_2, z_3 excluding its boundries.

Ans.

Sol. $\frac{\bar{z} - \bar{z}_1}{|z - z_1|^2} + \frac{\bar{z} - \bar{z}_2}{|z - z_2|^2} + \frac{\bar{z} - \bar{z}_3}{|z - z_3|^2} = 0$

$t_1(\bar{z} - \bar{z}_1) + t_2(\bar{z} - \bar{z}_2) + t_3(\bar{z} - \bar{z}_3) = 0$

where $|z - z_1|^2 = \frac{1}{t_1}$ etc



where $t_1, t_2, t_3 \in R^+$

$t_1(z - z_1) + t_2(z - z_2) + t_3(z - z_3) = 0$

$(t_1 + t_2 + t_3)\, z = t_1 z_1 + t_2 z_2 + t_3 z_3 \quad \Rightarrow \quad z = \dfrac{t_1 z_1 + t_2 z_2 + t_3 z_3}{t_1 + t_2 + t_3}$

$= \dfrac{t_1 z_1 + t_2 z_2}{t_1 + t_2} \cdot \dfrac{t_1 + t_2}{t_1 + t_2 + t_3} + \dfrac{t_3 z_3}{t_1 + t_2 + t_3} = \dfrac{t_1 + t_2}{t_1 + t_2 + t_3}\, z' + \dfrac{t_3 z_3}{t_1 + t_2 + t_3}$

$\Rightarrow \quad z = \dfrac{(t_1 + t_2) z' + t_3 z_3}{t_1 + t_2 + t_3} \quad \Rightarrow \quad z$ lies inside the $\Delta z_1 z_2 z_3$

If $t_1 = t_2 = t_3 \Rightarrow z$ is the centroid of the triangle . Also if $a = b = c$

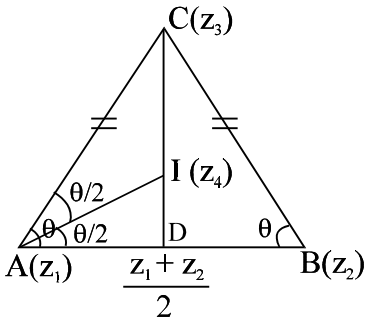
$\Rightarrow \quad |z - z_1| = |z - z_2| = |z - z_3| \Rightarrow z$ is the circumcentre]

Q.8 On the Argand plane z_1, z_2 and z_3 are respectively the vertices of an isosceles triangle ABC with AC = BC and equal angles are q. If z_4 is the incentre of the triangle then prove that $(z_2 - z_1)(z_3 - z_1) = (1 + \sec q)(z_4 - z_1)^2$

Ans.

Sol $\dfrac{z_2 - z_1}{|z_2 - z_1|} = \dfrac{z_4 - z_1}{|z_4 - z_1|} e^{-i\theta/2}$ ....(1) (clockwise)

and $\dfrac{z_3 - z_1}{|z_3 - z_1|} = \dfrac{z_4 - z_1}{|z_4 - z_1|} e^{i\theta/2}$ ....(2) (anticlockwise)



multiplying (1) and (2)

$\dfrac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)^2} = \dfrac{|(z_2 - z_1)|\,|(z_3 - z_1)|}{|z_4 - z_1|^2}$

$= \dfrac{(AB)|AC|}{(AI)^2} = \dfrac{2(AD)\,(AC)}{(AI)^2}$

$= \dfrac{2(AD)^2}{(AI)^2} \cdot \dfrac{AC}{AD} = 2\cos^2\dfrac{\theta}{2} \sec\theta = (1 + \cos\theta)\sec\theta$

5. DE-MOIVRE'S THEOREM

Case I If n is any integer then $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

Case II **Statement :** $n = \dfrac{p}{q}$; If $p, q \in Z$ and $q \neq 0$

Convert complex number into exponential form

Replace argument by general value of argument (Add $2k\pi$ to the argument) .

Apply given power

put $k = 0, 1, 2, \ldots, q - 1$

Q.1 $z = \left(1 + i\sqrt{3}\right)^{1/4}$

Ans. $2^{1/4}\left(e^{i\left(\frac{2k\pi + \pi/3}{4}\right)}\right)$; $k = 0, 1, 2, 3$

Sol. $z = \left(1 + i\sqrt{3}\right)^{1/4}$

$$z^4 = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$z^4 = 2\left\{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right\}$$

$$z^4 = 2\cdot\left\{e^{i\frac{\pi}{5}}\right\}$$

$$= 2e^{i\left(\frac{\pi}{3}+2k\pi\right)}$$

$$z = 2^{\frac{1}{4}}\cdot e^{i\left(\frac{\pi}{12}+\frac{2k\pi}{4}\right)}; k \in \{0,1,2,3\}$$

Q.2 $2\sqrt{2}\,z^4 = \left(\sqrt{3}-1\right) + i\left(\sqrt{3}+1\right)$

Ans. $z = e^{i\left(\frac{2k\pi+5\pi/12}{4}\right)}$; $k = 0,1,2,3$

Sol. $2\sqrt{2}\,z^4 = \left(\sqrt{3}-1\right) + i\left(\sqrt{3}+1\right)$

$$z^4 = \frac{\sqrt{3}-1}{2\sqrt{2}} + \frac{i\sqrt{3}+1}{2\sqrt{2}}$$

$$= \cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12} \qquad \left\{\frac{5\pi}{12} = 75^\circ\right\}$$

$$= e^{i\frac{5\pi}{12}}$$

$$= e^{i\left(2k\pi+\frac{5\pi}{12}\right)}$$

$$z = e^{i\frac{\left(2k\pi+\frac{5\pi}{12}\right)}{4}} \qquad k \in \{0,1,2,3\}$$

Q.3 Find the roots of the equation

(a) $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ (b) $z^4 - z^3 + z^2 - z + 1 = 0$

Ans. (a) $z = e^{i\frac{k\pi}{3}}$ $k = 1,2,3,4,5$ (b) $z = e^{i\left(\frac{(2k+1)\pi}{5}\right)}$ $k = 0,1,3,4$

Sol. (a) $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$

$$\frac{1\left(z^6-1\right)}{z-1} = 0$$

$$z^6 = 1; z \neq 1$$

$$z^6 = e^{i(2k\pi)}$$

$$z = e^{i\left[\frac{k\pi}{3}\right]}; k \neq 0$$

$k = 1, 2, 3, 4, 5$

(b) $z^4 - z^3 + z^2 - z + 1 = 0$

$$\frac{1\left((-z)^5 - 1\right)}{(-z) - 1} = 0$$

$-z^5 - 1 = 0$ & $z \pm -1$

$z^5 = -1$

$= e^{i\pi}$

$= e^{i(2k\pi + \pi)}$

$$z = e^{i\frac{(\pi + 2k\pi)}{5}} ; k \neq 2$$

$k = 0, 1, 3, 4$

Q.4 Find the number of roots of the equation $z^{10} - z^5 - 992 = 0$ with real part –ve.

Ans. 5

Sol. $z^{10} - z^5 - 992 = 0$

$z^{10} - 32\, z^5 + 31z^5 - 992 = 0$

$z^5 (z^5 - 32) + 31(z^5 - 32) = 0$

$(z^5 + 31)(z^5 - 32) = 0$

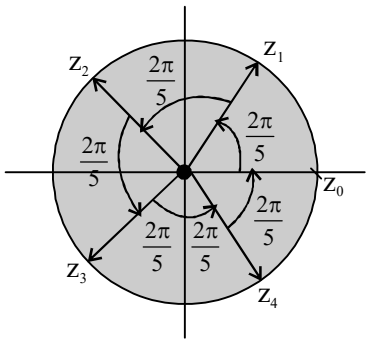
$\Rightarrow z^5 = -31$ or $z^5 = 32$

Now we neeal to find number of complex numbers with negative Real part.

$z^5 = 2^5$

with –ve real part

z_2 & z_3



& $z^5 = -31$

will be like mirror image of above diagram wrt y-axis but with magnitule $\left(31^{\frac{1}{5}}\right)$

So 3 roots will have –ve Real parts

$\therefore$ Total Roots with –ve Real Part 2 + 3 i.e 5.

Q.5 Prove that $\tan^{-1}\frac{1}{5} \approx \frac{\pi}{16}$

Ans.

Sol. Let $z = 5 + i$

$$\arg z = \tan^{-1}\frac{1}{5}$$

Now $z = 5 + i$

$z^2 = (5 + i)^2$

$= 25 - 1 + 10i$

$z^2 = 24 + 10i$

$z^4 = (24 + 10i)^2$

$= 576 - 100 + 480i$

$= 476 + 480i$

$\arg z^4 = \arg(476 + 480i)$

$$4 \arg z = \tan^{-1} \frac{480}{476}$$

$$\simeq \frac{\pi}{4}$$

$$\arg z \simeq \frac{\pi}{16}$$

6. CUBE ROOT OF UNITY

The cube roots of unity are 1, $\frac{-1 + i\sqrt{3}}{2} = \omega$, $\frac{-1 - i\sqrt{3}}{2} = \omega^2$.

Properties

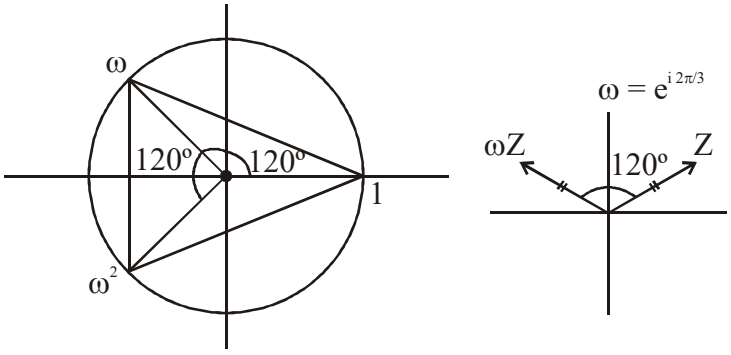
1. $Z^3 - 1 = 0$; Roots are 1, ω, ω^2

   $1 + \omega + \omega^2 = 0$; $\omega^3 = 1$; $\omega^{3k} = 1$, $k \in I$

2. $\overline{\omega} = \omega^2$; $|\omega| = 1$

3. $1 + \omega^r + \omega^{2r} \begin{cases} 3 & r \in 3k \\ 0 & r \notin 3k \end{cases}$

4. The three cube roots of unity when plotted on the argand plane constitute the verties of an equilateral triangle.



5. On multiplying any complex number z by ω it gets rotated by $\frac{2\pi}{3}$ about origin in anticlockwise direction
6. The following factorisation should be remembered :

   $x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$; $x^2 + x + 1 = (x - \omega)(x - \omega^2)$

   $x^3 + 1 = (x + 1)(x + \omega)(x + \omega^2)$; $x^2 - x + 1 = (x + \omega)(x + \omega^2)$

   $a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$; $a^2 + ab + b^2 = (a - b\omega)(a - b\omega^2)$;

$a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$; $a^2 - ab + b^2 = (a + b\omega)(a + b\omega^2)$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$

7. $-\omega = \frac{1}{2} - \frac{i\sqrt{3}}{2} = e^{-i\frac{\pi}{3}}$; $-\omega^2 = \frac{1}{2} + \frac{i\sqrt{3}}{2} = e^{i\frac{\pi}{3}}$

Q.1. Compute :

(i) $(2 - \omega)(2 - \omega^2)$ (ii) $(i + \omega)(i + \omega^2)$

Ans. (i) 7 (ii) $-i$

Sol. (i) As $x^2 + x + 1 = (x - \omega)(x - \omega^2)$

Put $x = 2$

$2^2 + 2 + 1 = (2 - \omega)(2 - \omega^2)$

$\therefore (2 - \omega)(2 - \omega^2) = 7$

(ii) As $x^2 - x + 1 = (x + \omega)(x + \omega^2)$

Put $x = i$

$i^2 + i + 1 = (i + \omega)(i + \omega^2)$

$\therefore (i + \omega)(i + \omega^2) = -i$

Q.2. If roots of equation $z^{2016} - 2z^3 + 2 = 0$ are $\lambda_1, \lambda_2, \ldots\ldots\ldots \lambda_{2016}$. Find value of $\prod_{i=1}^{2016}(\omega - \lambda_i)$.

Ans. 1

Sol. $z^{2016} - 2z^3 + 2 = 0 \rightarrow \lambda_1, \lambda_2, \vdots, \lambda_{2016}$

$\therefore z^{2016} - 2z^3 + 2 \equiv (z - \lambda_1)(z - \lambda_2).....(z - \lambda_{2016})$

put $z = \omega$

$\omega^{2016} - 2\omega^3 + 2 = (\omega - \lambda_1)(\omega - \lambda_2).....(\omega - \lambda_{2016})$

$1 - 2 + 2 = \prod_{i=1}^{2016}(\omega - \lambda_i)$

$\therefore \prod_{i=1}^{2016}(\omega - \lambda_i) = 1$

$i = 1$

Q.3. (i) If roots of equation $z^2 - z + 1 = 0$ are α, β find $\alpha^{2016} + \beta^{2016}$

(ii) Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to :

माना समीकरण $x^2 - \sqrt{2}x + 2 = 0$ के मूल α, β हैं। तो $\alpha^{14} + \beta^{14}$ बराबर है :

Ans. (i) 2 (ii) -128

Sol. (i) $z^2 - z + 1 = 0$

$z = \frac{1 \pm \sqrt{1 - 4}}{2}$

$= \frac{1+\sqrt{3}i}{2}, \frac{1-\sqrt{3}i}{2}$

Let $\alpha = \omega$ and $\beta = \omega^2$

Find $\omega^{2016} + (\omega^2)^{2016}$

$\Rightarrow 1 + 1$

$\Rightarrow 2$

(ii) $x^2 - \sqrt{2}x + 2 = 0$

$$x = \frac{\sqrt{2} \pm \sqrt{2-8}}{2} = \frac{\sqrt{2} \pm \sqrt{6}i}{2}$$

$$\alpha = \frac{\sqrt{2}+\sqrt{6}i}{2} = \sqrt{2}e^{\frac{i\pi}{3}} \ \& \ \beta = \sqrt{2}e^{\frac{-i\pi}{3}}$$

$$\alpha^{14} = 2^7 e^{\frac{i14\pi}{3}} = 128\left[e^{\frac{i2\pi}{3}}\right]$$

$$\beta^{14} = 128\left[e^{\frac{-i2\pi}{3}}\right]$$

$$\alpha^{14} + \beta^{14} = 128(2)\cos\left(\frac{2\pi}{3}\right) = -128$$

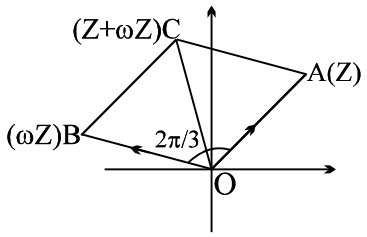
Q.4. If the area of the triangle in the Argand diagram, formed by Z, ωZ and Z + ωZ where ω is the imaginary cube root of unity is $16\sqrt{3}$ square units, then |Z| is

Ans.

Sol. Z, ωZ, Z + ωZ represent the vertices A, B, C of the parallelogram OACB in the Argand plane (wz is the vector obtaining by turning the vector z by 2π/3 anticlockwise.)

$\therefore$ area of $\Delta ABC = 1/2(OA)(OB)\sin\left(\frac{2\pi}{3}\right)$

$= 1/2\ |Z|^2\ \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}\ |Z|^2$



By data, $\frac{\sqrt{3}}{4}\ |Z|^2 = 16\ \sqrt{3} \Rightarrow |Z| = 8$

Q.5. Find all the complex numbers z satisfying $z^2 + z|z| + |z^2| = 0$

Ans.

Sol. $\left(\frac{z}{|z|}\right)^2 + \frac{z}{|z|} + 1 = 0$; $\alpha^2 + \alpha + 1 = 0$

$\Rightarrow \frac{z}{|z|} = \omega$ or $\omega^2 \Rightarrow$ $z = \lambda\omega, \lambda\omega^2$ where $\lambda > 0$

Q.6. If $\frac{a}{b} = \frac{b}{c} = \frac{c}{a} = k \neq 1$ find k.

Ans. $k = \omega\ ,\ \omega^2$

Sol. Now a = bk, b = ck, c = ak

$\therefore$ a = ck. k

a = ak . k . k

$k^3 = 1$

$\therefore$ k = 1, ω, ω^2 but $k \neq 1$

$\therefore$ $k = \omega$ & ω^2

Q.7. If $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$, then prove that z_1 , z_2 and z_3 are vertices of an equilateral triangle.

Ans.

Sol. If $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$ then $\dfrac{(z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_1)^2}{2} = 0$

(may simply $z_1 = z_2 = z_3$)

Also, $z_1^2 + z_2^2 + z_3^2 - z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$

$(z_1 + w z_2 + w^2 z_3)(z_1 + w^2 z_2 + w z_3) = 0$ (where w is the cube root of unity)

Let $z_1 + w z_2 + w^2 z_3 = 0$

$z_1 + w z_2 + (-1 - w) z_3 = 0$

$z_1 - z_3 = w (z_3 - z_2)$ ....(1)

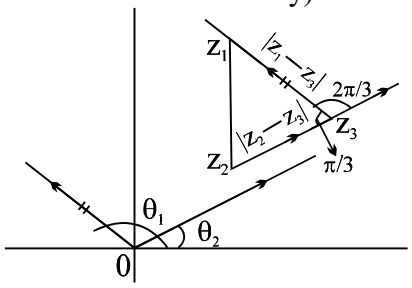
$\therefore$ $| z_1 - z_3 | = | w (z_3 - z_2) | = | w | | z_3 - z_2 |$

$| z_1 - z_3 | = | z_3 - z_2 |$

and amp $(z_1 - z_3)$ = amp w + amp $(z_3 - z_2)$

amp $(z_1 - z_3)$ – amp $(z_3 - z_2) = \dfrac{2\pi}{3}$

$\theta_1 - \theta_2 = \dfrac{2\pi}{3}$]



Q.8. If $(a+w)^{-1} + (b+w)^{-1} + (c+w)^{-1} + (d+w)^{-1} = 2\, w^{-1}$ and $(a+w^2)^{-1} + (b+w^2)^{-1} + (c+w^2)^{-1} + (d+w^2)^{-1} = 2\, w^{-2}$ where w is the imaginary cube root of unity then show that :

(i) Σ abc = 2 (ii) Σ a = 2 Π a .

(iii) $(a+1)^{-1} + (b+1)^{-1} + (c+1)^{-1} + (d+1)^{-1} = 2$, a, b, c, d $\in$ R.

Ans.

Sol. If $\dfrac{1}{a+\omega} + \dfrac{1}{b+\omega} + \dfrac{1}{c+\omega} + \dfrac{1}{d+\omega} = \dfrac{2}{\omega}$ ....(1)

and $\dfrac{1}{a+\omega^2} + \dfrac{1}{b+\omega^2} + \dfrac{1}{c+\omega^2} + \dfrac{1}{d+\omega^2} = \dfrac{2}{\omega^2}$ ....(2)

(1) and (2) implies that ω and ω^2 are the roots of the equation

$$\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} + \frac{1}{d+x} = \frac{2}{x}$$

$\Rightarrow$ $2x^4 + \left(\sum a\right)x^3 - \left(\sum abc\right)x - 2\prod a = 0$ ....(3)

$2\omega + \left(\sum a\right) - \left(\sum abc\right)\omega - 2\prod a = 0$ ....(4)

$2\omega^2 + \left(\sum a\right) - \left(\sum abc\right)\omega^2 - 2\prod a = 0$ ....(5)

subract – – + +

$2(\omega - \omega^2) - (\sum abc)(\omega - \omega^2) = 0 \quad \Rightarrow \quad (\sum abc) = 2$

$(\sum a) = 2\prod a$.

Also put x = 1 in equation (3) & verify that x = 1 satisfies it.

Hence $\frac{1}{a+1}+\frac{1}{b+1}+\frac{1}{c+1}+\frac{1}{d+1}=2$. Also the 4th root can be computed as

$1.\omega.\omega^2 r = -\frac{2abcd}{2} = -abcd \Rightarrow r = abcd$

Q.9. If $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha) = -3/2$ then prove that :

(a) $\Sigma \cos 2\alpha = 0 = \Sigma \sin 2\alpha$ (b) $\Sigma \sin(\alpha+\beta) = 0 = \Sigma \cos(\alpha+\beta)$

(c) $\Sigma \sin^2\alpha = \Sigma \cos^2\alpha = 3/2$ (d) $\Sigma \sin 3\alpha = 3\sin(\alpha+\beta+\gamma)$

(e) $\Sigma \cos 3\alpha = 3\cos(\alpha+\beta+\gamma)$ (f) $\sum\cos(2\alpha-\beta-\gamma)=3$, $\sum\sin(2\alpha-\beta-\gamma)=0$

7. n^{th} ROOT OF UNITY

$z^n = 1 \Rightarrow z^n = e^{i0} \Rightarrow z^n = e^{i2k\pi}$

$z = e^{i\left(\frac{2k\pi}{n}\right)}$

$k = 0 \quad e^{i0} = 1$

$k = 1 \quad e^{i\left(\frac{2\pi}{n}\right)} = \alpha_1$

$k = 2 \quad e^{i\left(\frac{4\pi}{n}\right)} = \alpha_2$

-

$k = n-1 \quad e^{i\left(\frac{2(n-1)\pi}{n}\right)} = \alpha_{n-1}$; Here $\alpha_K = e^{i\left(\frac{2k\pi}{n}\right)}$

Properties

1. $z^n - 1 = 0$; roots are $1, \alpha_1, \alpha_2, \ldots\ldots, \alpha_{n-1}$

(i) Sum of roots $= 0 \Rightarrow 1 + \alpha_1 + \alpha_2 + \ldots\ldots + \alpha_{n-1} = 0$

$\sum_{k=0}^{n-1} e^{i\left(\frac{2k\pi}{n}\right)} = 0 \Rightarrow \sum_{k=0}^{n-1}\cos\left(\frac{2k\pi}{n}\right)=0 \ \& \ \sum_{k=0}^{n-1}\sin\left(\frac{2k\pi}{n}\right)=0$

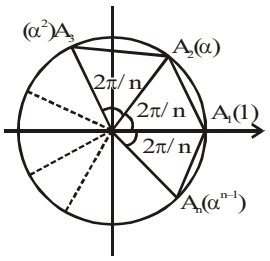
(ii) Product of roots $= S_n = -1$ if n = even; $S_n = 1$ if n = odd

2. $1, \alpha_1, \alpha_2, \alpha_3 \ldots\ldots \alpha_{n-1}$ are in G.P. with common ratio $e^{i(2\pi/n)}$

$|\alpha_k| = 1; \quad (\alpha_k)^n = 1; \quad \alpha_k \alpha_{n-k} = 1; \quad \alpha_{n-k} = \bar{\alpha}_k$

3. $x^n - 1 \equiv (x-1)(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_{n-1})$

4. $1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p$ = 0 if p is not an integral multiple of n
= n if p is an integral multiple of n



5. The points represented by n, n^{th} roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having center at origin, one vertex being on positive real axis.

Questions

Q.1 $\sum_{\lambda=1}^{12}\left(\sin\frac{2\lambda\pi}{13} - i\cos\frac{2\lambda\pi}{13}\right)$

Ans. **i**

Q.2 Find $(1-\alpha_1)(1-\alpha_2)..... (1-\alpha_{n-1})$

Ans. n

Q.3 Find $(1+\alpha_1)(1+\alpha_2)..... (1+\alpha_{n-1})$

Ans. 0 if n = even ; 1 if n = odd

Q.4 (a) Find $(i-\alpha_1)(i-\alpha_2)..... (i-\alpha_{n-1})$

Ans. o if n = 4k
1 if n = 4k + 1
1+i if n = 4k + 2
i if n = 4k + 3

Q.4 (b) Find $(\omega-\alpha_1)(\omega-\alpha_2)..... (\omega-\alpha_{n-1})$

Ans. o if n = 3k
1 if n = 3k + 1
$-\omega^2$ if n = 3k + 2

Q.5 If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to :

यदि समीकरण $x^4 + x^3 + x^2 + x + 1 = 0$ के मूल $\alpha, \beta, \gamma, \delta$ हैं, तो $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ बराबर हैं: :

Ans. –1

Sol. $x^4 + x^3 + x^2 + x + 1 = 0$ OR $\frac{x^5-1}{x-1} = 0 (x \neq 1)$

So roots are $e^{i2\pi/5}, e^{i4\pi/5}, e^{i6\pi/5}, e^{i8\pi/5}$

i.e. α, β, γ and δ

From properties of nth root of unity

$1^{2021} + \alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021} = 0$

$\Rightarrow \alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021} = -1$

Q.6 Factorise (z^7-1) into linear and quadratic factors and deduce that

(i) $\cos\frac{\pi}{7}.\cos\frac{2\pi}{7}.\cos\frac{3\pi}{7} = \frac{1}{8}$ (ii) $\sin\frac{\pi}{7}.\sin\frac{2\pi}{7}.\sin\frac{3\pi}{7} = \frac{\sqrt{7}}{8}$

Ans.

Sol $|1-\alpha_1|^2 |1-\alpha_2|^2 ..... |1-\alpha_{n-1}|^2 = n^2$ ....(1)

$(1-\alpha_1)(1-\overline{\alpha}_1)(1-\alpha_2)(1-\overline{\alpha}_2) .....(1-\alpha_{n-1})(1-\overline{\alpha}_{n-1}) = n^2$

consider $(1-\alpha_1)(1-\overline{\alpha}_1)$

where $\alpha_1 = \cos\theta + i\sin\theta$; $\overline{\alpha}_1 = \cos\theta - i\sin\theta$ and $\theta = \dfrac{2\pi}{n}$

$1 - (\alpha_1 + \overline{\alpha}_1) + \alpha_1\,\overline{\alpha}_1$

$1 - 2\cos\theta + 1 = 2(1 - \cos\theta) \Rightarrow 4\sin^2\theta/2$

$\therefore$ (1) becomes $2^{n-1}\sin\left(\dfrac{\theta}{2}\right).\sin 2\left(\dfrac{\theta}{2}\right).\sin 3\left(\dfrac{\theta}{2}\right)....\sin(n-1)\left(\dfrac{\theta}{2}\right) = n$ (where $\dfrac{\theta}{2} = \dfrac{\pi}{n}$)

when $n = 7$

$2^6 \sin\dfrac{\pi}{7}\ \sin\dfrac{2\pi}{7}\ \sin\dfrac{3\pi}{7}\ \sin\dfrac{4\pi}{7}\ \sin\dfrac{5\pi}{7}\ \sin\dfrac{6\pi}{7} = 7$

or $\sin^2\dfrac{\pi}{7}\ \sin^2\dfrac{2\pi}{7}\ \sin^2\dfrac{3\pi}{7} = \dfrac{7}{64}$ $\Rightarrow$ result]

Q.7 Prove that all roots of the equation $\left(\dfrac{z+1}{z}\right)^n = 1$ are collinear on the complex plane.

Q.8 If Z_r, $r = 1, 2, 3, ... 2m$, $m \in N$ are the roots of the equation

$Z^{2m} + Z^{2m-1} + Z^{2m-2} + ... + Z + 1 = 0$ then prove that $\sum\limits_{r=1}^{2m} \dfrac{1}{Z_r - 1} = -m$

Ans.

Sol

Method 1

Let $Z^{2m} + Z^{2m-1} + Z^{2m-2} + ... + Z + 1 \equiv (Z - Z_1)(Z - Z_2)...(Z - Z_{2m})$

Taking log on both the sides and diffrentiating w.r.t. Z

$$\frac{2mZ^{2m-1} + (2m-1)Z^{2m-2} + ... + 2z + 1}{Z^{2m} + Z^{2m-1} + Z^{2m-2} + ... + Z^2 + Z + 1} = \frac{1}{Z - Z_1} + \frac{1}{Z - Z_2} + ...+ \frac{1}{Z - Z_{2m}}$$

put $z = 1$, $\dfrac{1 + 2 + 3 + ... + 2m}{(2m+1)} = \dfrac{1}{1 - Z_1} + \dfrac{1}{1 - Z_2} + ...+ \dfrac{1}{1 - Z_{2m}}$

$\therefore$ $\displaystyle\sum_{r=1}^{2m} \frac{1}{Z_r - 1} = -\left[\frac{2m(2m+1)}{2(2m+1)}\right] = -m$]

Method 2

L.H.S. : note that $Z_1, Z_2, ..., Z_{2m}$ are the roots of $Z^{2m+1} = 1$,

now $\dfrac{1}{Z_1 - 1} + \dfrac{1}{Z_2 - 1} + \dfrac{1}{Z_3 - 1} + ... + \dfrac{1}{Z_{2m} - 1}$

Clubbing 1st and last ; 2nd and 2nd last and so on

$$= \left(\frac{1}{Z_1 - 1} + \frac{1}{Z_{2m} - 1}\right) + \left(\frac{1}{Z_2 - 1} + \frac{1}{Z_{2m-1} - 1}\right) + ... = (-1) + (-1) + .. \text{ m times} = -m$$

8.1 SUMMATION OF SERIES USING COMPLEX NUMBERS

(i) $\cos\theta + \cos 2\theta + \cos 3\theta + ... + \cos n\theta = \dfrac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\dfrac{n+1}{2}\right)\theta.$

(ii) $\sin\theta + \sin 2\theta + \sin 3\theta + ... + \sin n\theta = \dfrac{\sin(n\theta/2)}{\sin(\theta/2)} \sin\left(\dfrac{n+1}{2}\right)\theta.$

8.2 COMPLEX NUMBERS AND BINOMIAL COEFFICIENTS

(i) $C_0 + C_4 + C_8 + ...$

Ans. $\dfrac{2^{n-1} + 2^{n/2}\cos\left(\dfrac{n\pi}{4}\right)}{2}$

(ii) $C_1 + C_5 + C_9 + ...$

Ans. $\dfrac{2^{n-1} + 2^{n/2}\sin\left(\dfrac{n\pi}{4}\right)}{2}$

(iii) $C_2 + C_6 + C_{10} + ...$

Ans. $\dfrac{2^{n-1} - 2^{n/2}\cos\left(\dfrac{n\pi}{4}\right)}{2}$

(iv) $C_3 + C_7 + C_{11} + ...$

Ans. $\dfrac{2^{n-1} - 2^{n/2}\sin\left(\dfrac{n\pi}{4}\right)}{2}$

(v) $C_0 + C_3 + C_6 + C_9 + ...$

Ans. $\dfrac{2^n + (-\omega)^n + (-\omega^2)^n}{3}$

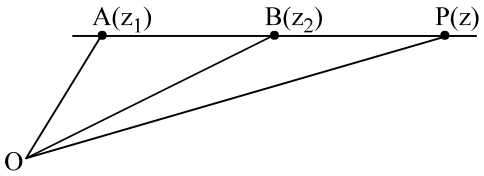
(vi) Prove that $\sum_{r=1}^{k} (-3)^{r-1} \cdot {}^{3n}C_{2r-1} = 0$ where $k = \dfrac{3n}{2}$, n is an even integer.

Sol. Consider the binomial expansion of $(1+i\sqrt{3})^{3n}$ and also using DMT as $2^{3n}\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)^{3n}$. Equate real and imaginary part.

9.1 STRAIGHT LINE IN COMPLEX NUMBER

1. Parametric Form

$\therefore \quad \dfrac{z - z_1}{z_2 - z_1}$ is purely real

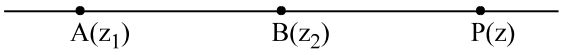


$z = z_1 + \lambda (z_2 - z_1)$

where $\lambda \in R$

2. Determinant Form

If $\vec{PA}$ & $\vec{PB}$ are always collinear i.e.



i.e. $\dfrac{z - z_1}{z - z_2}$ is purely real

i.e. $\dfrac{z - z_1}{z - z_2} = \dfrac{\bar{z} - \bar{z}_1}{\bar{z} - \bar{z}_2}$

$z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + (z_1 \bar{z}_2 - z_2 \bar{z}_1) = 0$

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

Note:

z_1, z_2 & z_3 are collinear if, $\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0$

3. General Form

$z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + (z_1 \bar{z}_2 - z_2 \bar{z}_1) = 0$

$z_1 \bar{z}_2$ & $z_2 \bar{z}_1$ are conjugate of each other $\Rightarrow$ $z_1 \bar{z}_2 - z_2 \bar{z}_1$ = purely imaginary

multiply equation by i

$i(\bar{z}_1 - \bar{z}_2) z - i(z_1 - z_2)\bar{z} + i(z_1 \bar{z}_2 - z_2 \bar{z}_1) = 0$

Let $-i(z_1 - z_2) = \alpha$; $\quad i(\bar{z}_1 - \bar{z}_2) = \bar{\alpha}$

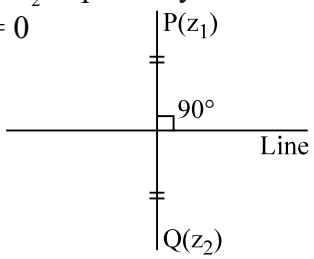
This reduces to $\bar{\alpha} z + \alpha \bar{z} + r = 0$ where $r = i(z_1 \bar{z}_2 - z_2 \bar{z}_1)$ = purely real

The equation $\bar{\alpha} z + \alpha \bar{z} + r = 0$ is also identified as the general equation of a line on complex plane.

Image of a point in a line

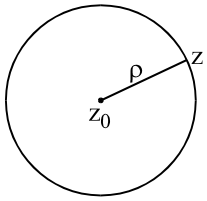
Two given points P & Q denoted by complex numbers z_1 and z_2 respectively are the reflection points in a given straight $\bar{\alpha} z + \alpha \bar{z} + r = 0$ if the given line is the right bisector of the line segment PQ.

Two points $P(z_1)$ & $Q(z_2)$ will be the reflection points of each other in the given straight line, iff $\bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$ where 'r' is real and α is non zero complex constant.



9.2 CIRCLE IN COMPLEX NUMBER

1. $|z - z_0| = \rho$ where ρ is the radius of the circle and z_0 is the centre



2. **General Form**

$|z - z_0| = \rho \quad \Rightarrow \quad |z - z_0|^2 = \rho^2 \quad \Rightarrow \quad (z - z_0)\ (\bar{z} - \bar{z}_0) = \rho^2$

$z\bar{z} - z\bar{z}_0 - z_0\bar{z} + z_0\bar{z}_0 - \rho^2 = 0$

if $-z_0 = \alpha$ then $-\bar{z}_0 = \bar{\alpha}$ & $z_0\bar{z}_0 = \alpha\bar{\alpha}$

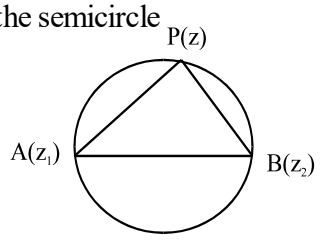
$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + \alpha\bar{\alpha} - \rho^2 = 0$

$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0 \quad$ — (1)

where $\alpha\bar{\alpha} - \rho^2 = r$ or $\rho = \sqrt{\alpha\bar{\alpha} - r}$

3. Diametrical Form

(i) Now let us consider a circle described on a line segment AB, $(A(z_1), B(z_2))$ as its diameter. Let P(z) be any point on the circle. As the angle in the semicircle is $\frac{\pi}{2}$, $\angle APB = \frac{\pi}{2}$



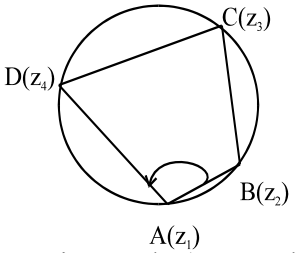
$\Rightarrow \arg\left(\frac{z_1 - z}{z_2 - z}\right) = \pm\frac{\pi}{2} \Rightarrow \frac{z - z_1}{z - z_2}$ is purely imaginary

$\Rightarrow \frac{z - z_1}{z - z_2} + \frac{\bar{z} - \bar{z}_1}{\bar{z} - \bar{z}_2} = 0 \Rightarrow (z - z_1)\ (\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$

(ii) $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$

Note: Let ABCD be a cyclic quadrilateral such that $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ lie on a circle. Clearly $\angle A + \angle C = \pi$

$\Rightarrow \arg\left(\frac{z_4 - z_1}{z_2 - z_1}\right) + \arg\left(\frac{z_2 - z_3}{z_4 - z_3}\right) = \pi$



$\Rightarrow \frac{z_4 - z_1}{z_2 - z_1} \cdot \frac{z_2 - z_3}{z_4 - z_3}$ is purely real.

Thus points $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ (taken in order) would be concyclic if $\frac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)}$ is purely real.